Antimalarial Agents
Chemoprophylaxis and Treatment of Malaria

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Following completion of the learning materials, students should be able to:

1. Define the terms exoerythrocytic stage, erythrocytic stage, schizont stage, schizogony, merozoite, latent liver stage, schizonticide, gametocide, hypnozoite, and hemozoin.
2. Differentiate the clinical approaches to malaria prophylaxis and treatment.
3. Apply the clinical selection of the various antimalarial drugs to their mechanisms of action, mechanisms of resistance, and geographical resistance patterns.
4. Relate the adverse effects of the antimalarial drugs to their therapeutic effects and give examples.
5. Explain the mechanism of toxicity and the effects of the drugs to be avoided in patients with glucose-6-phosphate dehydrogenase deficiency.
6. List the clinically relevant pharmacokinetic properties of the various chemotherapeutic agents used in the treatment of malarial infections and give examples.

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Preparation Materials (links are in the CPG and on the next slide)

Required

- ScholarRx Bricks | Practice Questions (questions FOR learning)

Highly relevant optional materials:

- Video Lecture | Dr. Goldstein's Word handout | Guided Reading Questions
- Textbook resources with links are listed on the next slide

SUGGESTIONS:

- Use the resources that work best for you.
- You do not need to study all of them.
- Work through the GUIDED READING QUESTIONS with pen/pencil and paper.

Try answering them in your OWN WORDS first without looking at the readings, and then again after reviewing.

Links

Name of Discipline/ Organ System: General Microbiology > Antiparasitic Drugs

Title of Brick: Drugs to treat protozoal and helminthic infections – the section on Malaria

Link: https://exchange.scholarrx.com/brick/drugs-to-treat-protozoal-and-helminthic-infections

Access Medicine Katzung's Basic & Clinical Pharmacology, 16e, 2024; Chapter 52: Antiprotozoal Drugs > Malaria

https://accessmedicine-mhmedical-com.nyit.idm.oclc.org/content.aspx?bookid=3382&sectionid=281755910#1204144707

The review books include practice questions in their chapters.

LWW Health Library, Premium Basic Sciences: Lippincott's Illustrated Reviews: Pharmacology 8e, 2023; Chapter 35 Antiprotozoal Drugs > Malaria https://premiumbasicsciences-lwwhealthlibrary-com.nyit.idm.oclc.org/content.aspx?sectionid=253329448&bookid=3222

Access Medicine Katzung's Pharmacology: Examination & Board Review, 1e, 2024; Chapter 52: Antiprotozoal Drugs > Malaria

https://nyit.idm.oclc.org/login?url=https://accessmedicine.mhmedical.com/content.aspx?bookid=3461&sectionid=285599695#1207306375

Outline

1. Overview of malaria-endemic countries
2. Plasmodium life cycle
3. Clinical features of malaria
4. Goals of therapy
5. Prophylaxis and treatment of malaria
6. Pharmacology of the individual antimalarial drugs
7. Check your knowledge questions
8. Summary

Here are the TAKEAWAYS of malaria prophylaxis and treatment

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- Malaria is caused by *Plasmodium* protozoan parasites transmitted by the bite of Anopheles mosqitoes. *P. falciparum* is the most widespread and deadliest malaria parasite.
- Prevention measures include personal protection using insect repellent, wearing long sleeves and pants, and sleeping under an insecticide treated bed net.
- Chemoprophylaxis is based on travel destination, duration of stay, and individual health considerations. Long-acting agents are used.
- Chloroquine is preferred for prophylaxis in infections caused by non-falciparum species and in regions without *P. falciparum* resistance. However, *P. falciparum* resistance is widespread.
- Atovaquone-proguanil, doxycycline, and mefloquine are used in chloroquine-resistance areas.
- Treatment involves combining a rapid-acting drug with a short duration of action and a long-acting drug with a longer half-life to provide extended duration of activity. The combination is also intended to reduce the development of parasite resistance to the drug therapy.
- Preferred treatment for uncomplicated *P. falciparum* malaria, and malaria caused by other *Plasmodium* species, is oral artemisinin combination therapy (ACT). ACT treatment duration is 3 days (in the absence of partial resistance to the artemisinins). Chloroquine may be used for treatment of chloroquine-sensitive malaria. Preferred treatment of severe *P. falciparum* malaria, which can be fatal, is parenteral artesunate (an artemisinin).
- Treatment of pregnant and pediatric patients follows the same principles and utilizes antimalarial drugs with evidence of safety and efficacy for these populations.
- *P. vivax* and *P. ovale* remain dormant in the liver as hypnozoites and can reactivate weeks, months, or years after initial infection, causing relapse. Primaquine and tafenoquine erradicate hypnozoites. Patients should be screened for G6PD deficiency prior to administration of these drugs.

Key points: Malaria and Clinical Management Overview

- Malaria is an acute infectious disease. Five species of the protozoa *Plasmodium* cause human malaria, which is transmitted to humans by the female Anopheles mosquito. Travelers to malarious regions and young children in endemic regions are at highest risk for severe disease.
- *P. falciparum* is the primary cause of severe malaria, an acute, rapidly progressive severe illness characterized by persistent high fever, hyperparasitemia, cerebral symptoms, organ system dysfunction, and death.
- *P. vivax*, *P. ovale*, *P. malariae*, and *P. knowlesi* can also cause significant illness. *P. vivax* can cause severe anemia. *P. ovale* can also cause anemia and splenomegaly. *P. vivax* and *P. ovale* species remain dormant in the liver (hypnozoite stage) and cause relapse weeks, months, or years later. *P. malariae* typically results in chronic low level parasitemia, but can cause severe anemia and other complications. *P. knowlesi*, an uncommon form of malaria in Southeast Asia previously thought to infect only nonhuman primates (macaques), causes human infections, sometimes severe with complications similar to those caused by *P. falciparum*.

Clinical management of malaria depends on the severity, age of the patient, presence or absence of acquired immunity, drug resistance patterns, and availability of drugs.

Antimalarial drugs are used for prevention of malaria and its complications in vulnerable populations, such as non-immune travelers, infants and children at high risk of severe disease, pregnant people, and for treatment of uncomplicated and severe malaria.

Antimalarial drugs can be classified based on their activities during the *Plasmodium* life cycle as well as by their intended use for either chemoprophylaxis or treatment.

Key points: Chemoprophylaxis and Treatment Overview

Chemoprophylaxis: *Drugs can only prevent the development of symptomatic malaria caused by the asexual erythrocytic forms*.

In other words, preventive use of antimalarial drugs is intended to kill the asexual erythrocytic parasites before they increase sufficiently in number to cause clinical disease.

Treatment of established infection: No single antimalarial is effective against the erythrocytic stages and liver stages of the life cycle that may co-exist in the same patient.

In other words, no single antimalarial drug can effect a **radical cure** (eliminate both latent hepatic and erythrocytic stages of *P. vivax* and *P. ovale*).

⇒ *More than one drug is required to completely eliminate both the erythrocytic and liver infection*.

- Patients on malaria prophylaxis who develop malaria infection should receive a different medication for treatment.

Blood schizonticides commonly used for chemoprophylaxis include chloroquine (chloroquine-sensitive regions), atovaquone-proguanil, and mefloquine.

For treatment of uncomplicated and severe malaria, the artemisinins partnered with a long-acting drug such as lumefantrine, mefloquine, or atovaquone-proguanil are first line. Quinine partnered with doxycycline or tetracycline is an alternative treatment regimen. Chloroquine is preferred for non-falciparum malaria.

Primaquine and tafenoquine are effective against the dormant hypnozoites of *P. vivax* and *P. ovale.*

Definitions

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Exoerythrocytic stage	the phase of the life cycle of malaria parasites in the liver cells of the host before the malaria parasites invade the red blood cells. The parasites develop and multiply within the liver during this stage.
Erythrocytic stage	the phase in the life cycle of malaria parasites where they infect and multiply within red blood cells (erythrocytes) of the host
Schizont stage Schizogony Merozoites	the phase in the life cycle when the parasite undergoes asexual reproduction (schizogony) within hepatocytes after the initial infection and then in red blood cells. The parasite nucleus divides multiple times, producing thousands of daughter cells (merozoites), which invade red blood cells.
Latent liver stage	a dormant phase of the malaria parasites *P. vivax* and *P. ovale* in the liver, where they remain inactive for a period before potentially causing a relapse of the disease
Schizonticide	a drug that specifically targets and destroys the schizont stage of a sporozoan parasite, such as those causing malaria
Gametocide	a drug that destroys the gametocytes (sexual forms) of malaria parasites, preventing them from being transmitted to mosquitoes
Hypnozoite	a dormant form of *P. vivax* and *P. ovale* in the liver, which can cause relapses in infections
Hemozoin	iron-containing pigment that is a breakdown product of hemoglobin. Malaria parasites digest hemoglobin releasing free heme, which is toxic to cells. The parasites convert the free heme to hemozoin, an insoluble crystalline form that is not toxic to the malaria parasite.

Geographic distribution of malaria parasites:

P. falciparum:
sub-Saharan Africa, South America, Southeast Asia, India – the most widespread and deadliest malaria parasite.

P. vivax:
Asia, Latin America, some parts of Africa

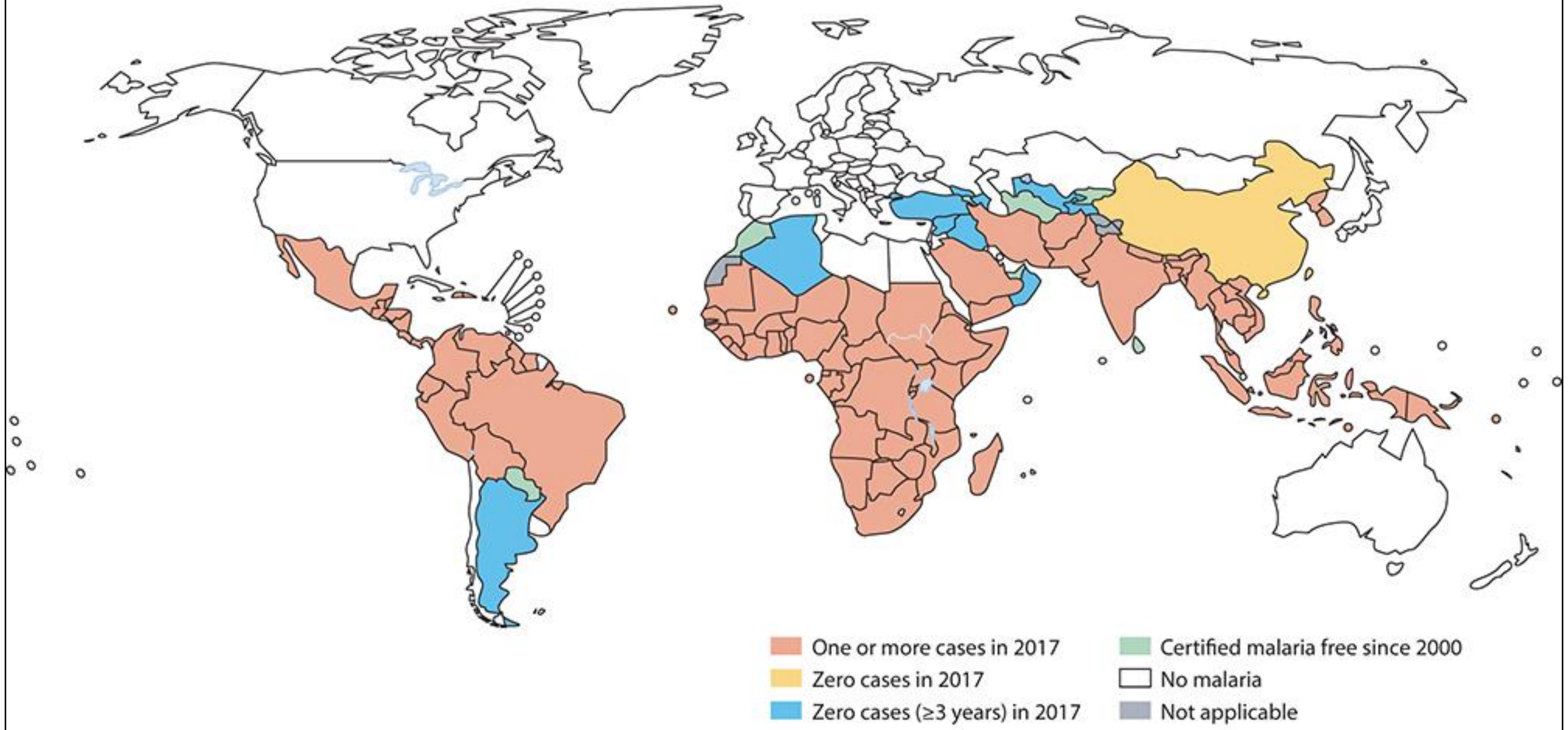
P. ovale:
West Africa primarily

P. malariae:
sub-Saharan Africa, Southeast Asia, Indonesia, parts of the Amazon Basin

P. knowlesi:
Southeast Asia (Malaysia, Thailand, Vietnam, Indonesia)

transmitted via the bite of anopheles mosquito from macaques to humans

Countries with indigenous cases in 2000 and their status by 2017 Countries with zero indigenous cases over at least the past 3 consecultive years are considered to be malaria free. All countries in the WHO European Region reported zero inidigenous cases in 2016 and again in 2017. In 2017, both China and El Salvador reportes zero indigenous cases. *Source: WHO database.*

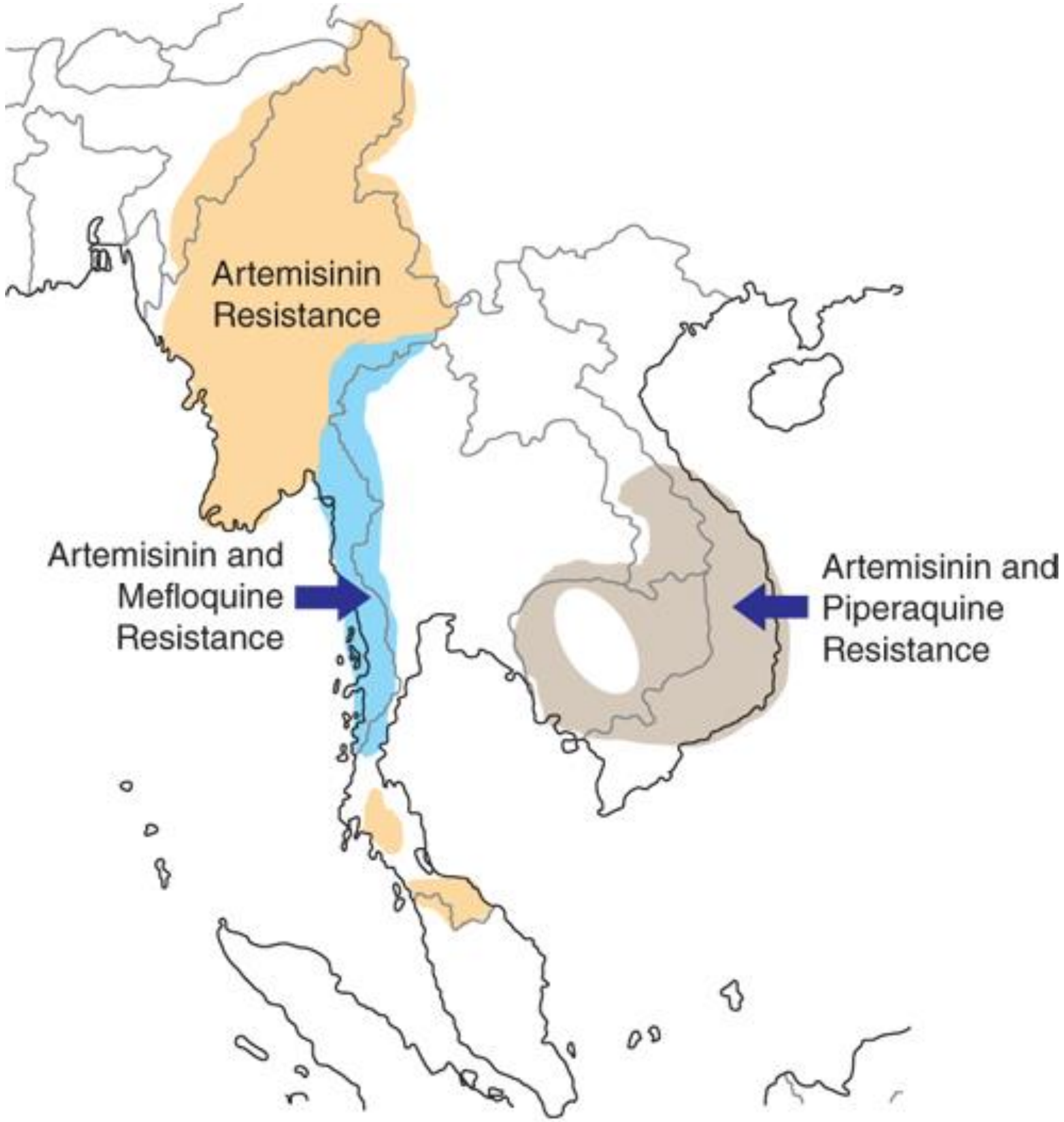


WHO:World Health Organization.

Source: Matthew L. Boulton, Robert B. Wallace: *Maxcy-Rosenau-Last Public Health & Preventive Medicine, 16e*

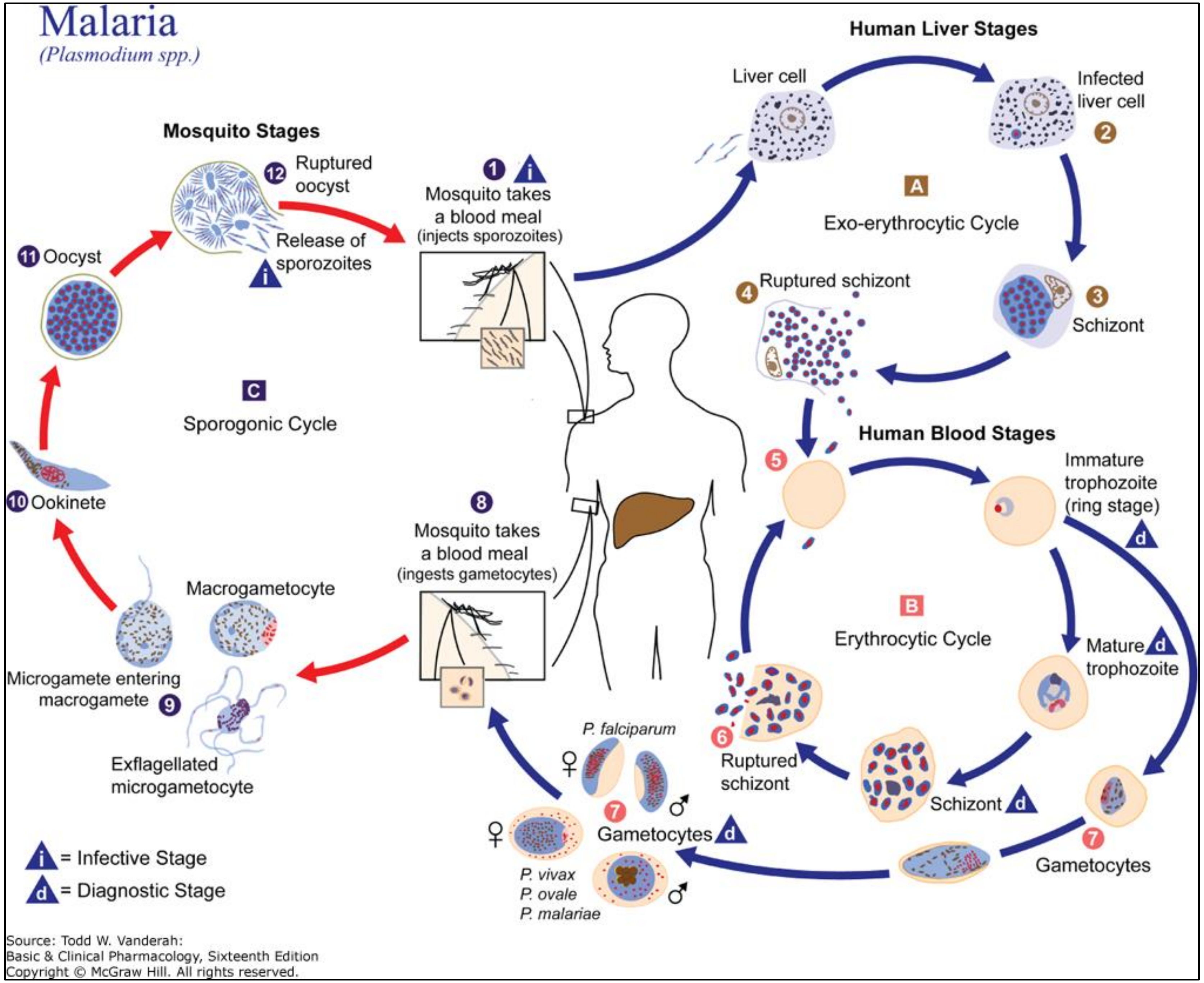
Global distribution of malaria in 2017.[1] (Source: Reprinted with permission of the World Health Organization.)

Citation: Chapter 89 Malaria, Boulton ML, Wallace RB. *Maxcy-Rosenau-Last Public Health & Preventive Medicine, 16e;* 2022. Available at: https://accessmedicine.mhmedical.com/content.aspx?sectionid=257463282&bookid=3078&Resultclick=2 Accessed: March 24, 2024



Current geographic extent of artemisinin resistance and artemisinin-based combination therapy partner drug resistance in *Plasmodium falciparum* in the Greater Mekong subregion.

Citation: Chapter 224 Malaria, Loscalzo J, Fauci A, Kasper D, Hauser S, Longo D, Jameson J. *Harrison's Principles of Internal Medicine, 21e;* 2022. Available at: https://accessmedicine.mhmedical.com/content.aspx?sectionid=265435854&bookid=3095&Resultclick=2 Accessed: March 24, 2024



- Sporozoites rapidly invade liver parenchymal cells and begin asexual reproduction → exoerythrocytic stage schizonts
- The schizonts rupture releasing numerous merozoites (asexual progeny) that enter the bloodstream, and invade red cells, and become trophozoites (ring stage). Some mature into schizonts.

  By the end of the erythrocytic stage, the parasite has consumed 2/3 of the RBC’s hemoglobin and occupies most of the cell.
- The schizonts rupture, releasing 6-30 merozoites which can invade more RBCs, repeating the cycle.
- Some merozoites differentiate into male and female gametocytes.

Only the asexual erythrocytic stage causes clinical malaria.

Life cycle of malaria parasites. Only the asexual erythrocytic stage of infection causes clinical malaria. All effective antimalarial treatments are blood schizonticides that kill this stage. (CDC/Alexander J. da Silva, PhD; Melanie Moser.)

Slide adapted by LG

Citation: Chapter 52 Antiprotozoal Drugs, Vanderah TW. *Katzung’s Basic & Clinical Pharmacology, 16th Edition;* 2024, figure 52-1.
Available at: https://accessmedicine.mhmedical.com/content.aspx?bookid=3382&sectionid=281755910 Accessed: March 25, 2024

Clinical Features of *P. falciparum* Malaria

Presentation can vary substantially depending on the infecting species, the level of parasitemia, the immune status of the patient, and age.

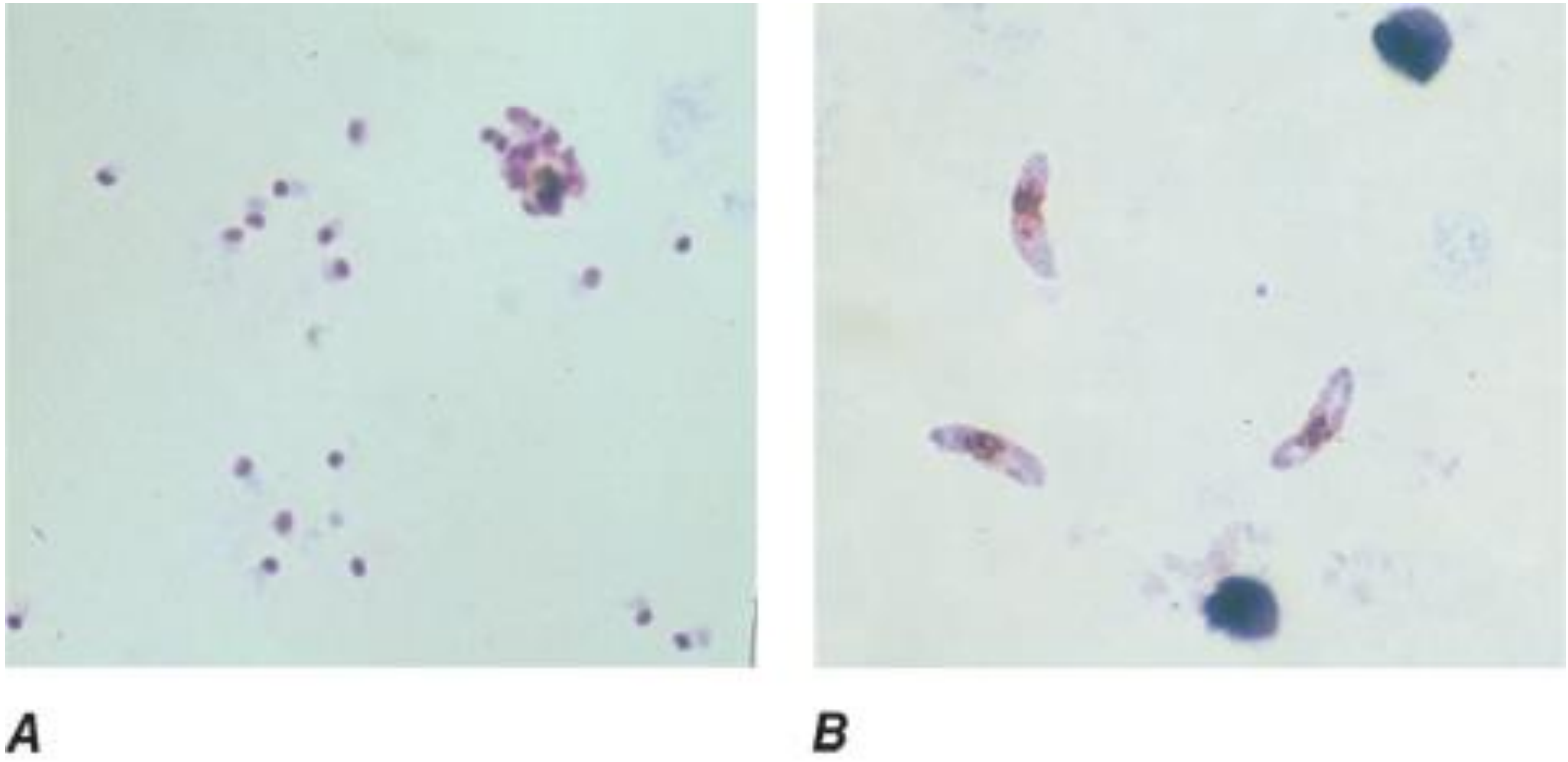
Signs and Symptoms	**Uncomplicated:** fever, malaise, mild anemia, palpable spleen in some cases
Complications of malaria	**Severe falciparum malaria:** Can be fatal hyperparasitemia, hemolysis, severe anemia, seizures hypotension/shock, hypoglycemia, acute kidney injury, coma acidosis, pulmonary edema/ARDS;
Lab	Intra-erythrocytic parasites identified in Giemsa-stained thick and thin blood smears

High-power view of a normal peripheral blood smear. Several platelets (arrows) and a normal lymphocyte (arrowhead) can also be seen. The red cells are of relatively uniform size and shape. The diameter of the normal red cell should approximate that of the nucleus of the small lymphocyte; central pallor (dashed arrow) should equal one-third of its diameter.

Bursting *Plasmodium*-infected red blood cell

https://www.homelandsecuritynewswire.com/promise-vaccine-against-deadly-malaria-parasite

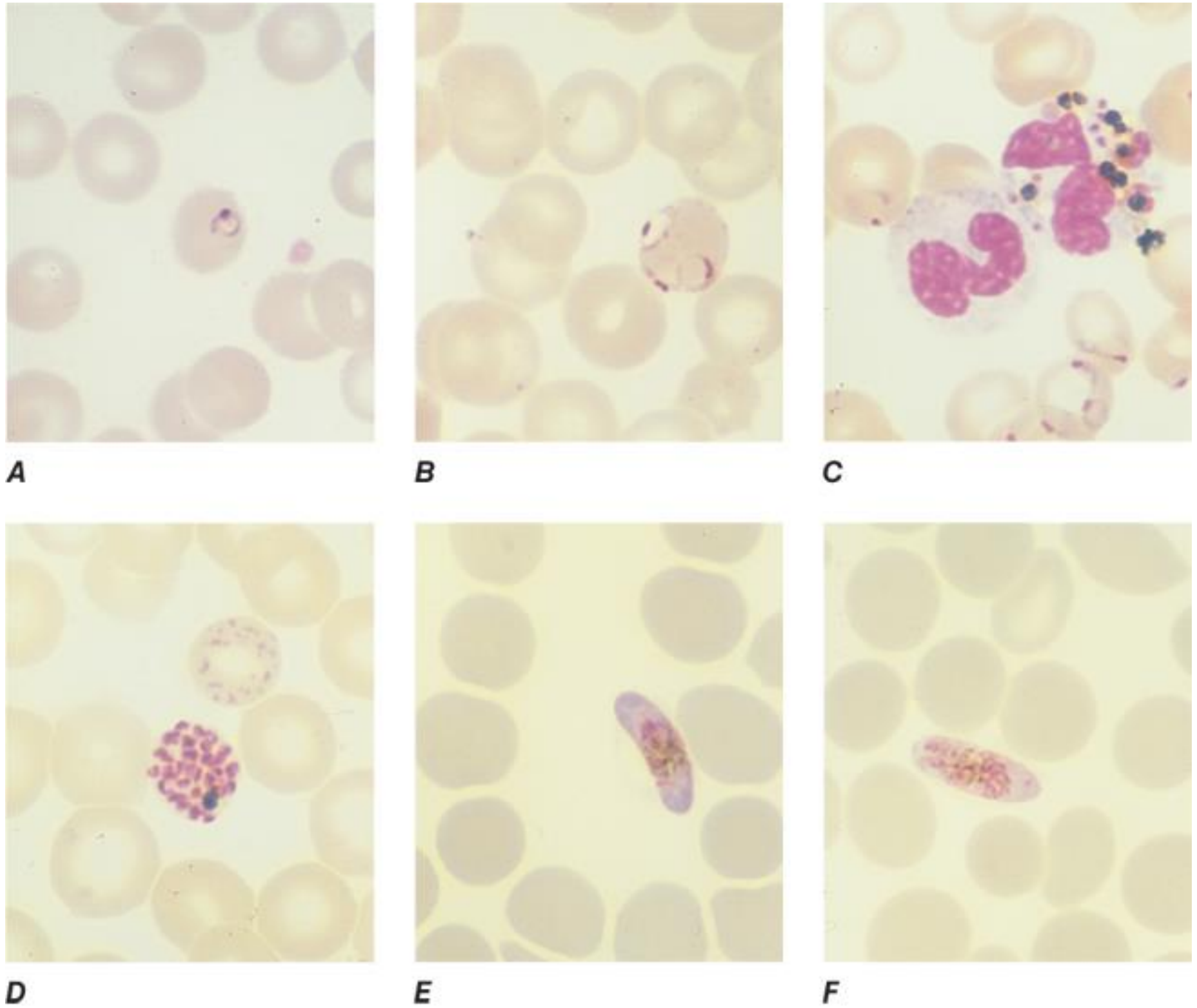
Harrison’s updated figure 219-6



Source: J.L. Jameson, A.S. Fauci, D.L. Kasper, S.L. Hauser, D.L. Longo, J. Loscalzo: Harrison's Principles of Internal Medicine, 20th Edition

Thick blood films of Plasmodium falciparum. A. Trophozoites. B. Gametocytes. (Reproduced from Bench Aids for the Diagnosis of Malaria Infections, 2nd ed, with the permission of the World Health Organization.)

Harrison’s updated figure 219-4



Source: J.L. Jameson, A.S. Fauci, D.L. Kasper, S.L. Hauser, D.L. Longo, J. Loscalzo: Harrison's Principles of Internal Medicine, 20th Edition

Thin blood films of Plasmodium falciparum. A. Young trophozoite. B. Old trophozoite. C. Trophozoites in erythrocytes and pigment in polymorphonuclear cells. D. Mature schizont. E. Female gametocyte. F. Male gametocyte. (Reproduced from Bench Aids for the Diagnosis of Malaria Infections, 2nd ed, with the permission of the World Health Organization.)

Chemoprophylaxis and Chemotherapy of Malarial Infections

Goals

1. Prevention of malaria in nonimmune travelers
2. Prevention of malaria and its complications in endemic regions in populations at high risk for severe disease, such as infants, children, and pregnant persons
3. Treatment of uncomplicated malaria
4. Treatment of severe malaria

Chemoprophylaxis:

- *The antimalarial drugs can prevent the development of symptomatic malaria caused by the* ***asexual erythrocytic forms***.

Treatment:

- Artemisinins in combination with a long-acting agent (ACT) are preferred for treatment.
- Chloroquine for chloroquine-sensitive malaria (or ACT).
- No one agent is effective against both latent hepatic and erythrocytic stages of *P. vivax* and *P. ovale.*
- A schizonticide plus primaquine or tafenoquine are required to effect radical cure (prevention of relapse).

Antimalarial drugs

Malaria Prophylaxis and Treatment

Long-acting oral drugs:

- ✳ Chloroquine
- ✳ Mefloquine
- ✳ Sulfadoxine-pyrimethamine (SP)
- ✳ Lumefantrine

  Only in fixed combination with artemisinin

Shorter-acting oral drugs

(more frequent dosing)

- ✳ Atovaquone-progruanil
- ✳ Doxycycline

Long-acting antimalarials not available in the U.S.
Amodiaquine
Piperaquine
Pyonardine tetraphosphate
Quinine I.V.

Malaria Treatment

Short-acting drugs:

- ✳ Arteminisin
- ✳ Artemether
- ✳ Artesunate (parenteral only)
- Dihydroartemisinin (not available in U.S.)
- ✳ Quinine

Slow-acting antimalarial drugs:

- ✳ Doxycycline
- ✳ Clindamycin

CATEGORIES OF ANTIMALARIAL AGENTS

EFFECT OF DRUG ON PARASITE VIABILITY

			LIVER STAGES		BLOOD STAGES	
GROUP	**DRUGS**	**SPOROZOITE**	**PRIMARY**	**HYPNOZOITE**	**ASEXUAL**	**GAMETOCYTE**
1	Artemisinins	–	–	–	+	+
	Chloroquine	–	–	–	+	+/–
	Mefloquine	–	–	–	+	–
	Quinine/Quinidine	–	–	–	+	+/–
	Pyrimethamine	–	–	–	+	–
	Sulfadoxine	–	–	–	+	–
	Tetracycline	–	–	–	+	–
2	Atovaquone – Proguanil (combo)	–	+	–	+	+/–
3	Primaquine	–	+	+	–	+
	Tafenoquine	+	+	+	+	+

+/–: not reliably effective at eliminating gametocytes

Goodman & Gilman's The Pharmacological Basis of Therapeutics, 14e, 2023; Table 66-1

Chemoprophylaxis in malaria endemic regions

drug	frequency	start	stop	Use in pregnancy	C-resistant *P. Falciparum*	C-sensitive *P. Falciparum*	*P. vivax*
Chloroquine	once weekly	1-2 weeks	4 weeks	yes		✓	✓
A-P	once daily	1-2 days	7 days	no	✓	✓	✓
Mefloquine	once weekly	2-3 weeks	4 weeks	yes	✓	✓	✓
Doxycycline	once daily	1-2 days	4 weeks	no	✓	✓	✓
Tafenoquine	once weekly	see below	see below	no	✓	✓	✓
Primaquine	once daily	1-2 days	7 days	no			✓

A-P: Atovaquone-proguanil

C: Chloroquine

Tafenoquine: A loading dose is taken 3 days before exposure, then once weekly starting 7 days after the loading dose.
Tafenoquine is approved for prophylaxis in people ≥18 years old and ≥16 years old to prevent relapse of *P. vivax*.

Primaquine: Appropriate for short-duration travel to areas with principally *P. vivax*.
Primaquine once daily for 14 days is also to prevent relapse from presumptive *P. vivax* or *P. ovale*.

Pediatrics: Chloroquine, atovaquone-proguanil, mefloquine, doxycycline (≥8 years old), and primaquine may be used.

Malaria Treatment (nonpregnant adults and pediatric patients) General guidelines (WHO 2022)

TAKEAWAYS: 1) Artemisinin combination therapy (ACT) is preferred. 2) Why: Rapid-acting and short-acting artemisinin is combined with long-acting drug. 3) Monitor parasite density.

Uncomplicated *P. falciparum* malaria		
Artemether-lumefantrine	orally, twice daily x3 days with full meal or milk	Patients >65 kg: consider an alternative agent Efficacy diminishes with increasing body weight.
Combinations available outside the U.S.; all are given orally once daily for 3 days. Artesunate-amodiaquine, artesunate-mefloquine, artesunate with sulfadoxine-pyrimethamine, dihydroartemisinin-piperaquine, and artesunate pyonardine tetraphosphate		

Severe *P. falciparum* malaria with no established artermisinin resistance		
Preferred	***Alternatives if artesunate is unavailable***	
Artesunate IV or IM at 0,12, and 24 hours then 1x daily until parasite density <1%, then oral ACT	In United States: Orally or via nasogastric tube until artesunate is available	Outside U.S.: 1) Artemether IM until artesunate is available or oral treatment can be given, *or* 2) Quinine IV until artesunate arrives
Severe *P. falciparum* malaria with established artemisinin resistance		
Artesunate IV or IM at 0,12, and 24 hours then 1x daily until parasite density <1%, then oral ACT	In United States: Orally or via nasogastric tube until artesunate is available	Outside U.S.: 1) Artemether IM until artesunate available or oral tolerated *plus* 2) Quinine IV until artesunate arrives

Pregnant Persons and Children: Chemoprophylaxis and Treatment

<table>
<tr><td colspan="3">Chemoprophylaxis for pregnant people</td></tr>
<tr><td>Travelers</td><td colspan="2">➢ Defer travel if possible. Pregnant people are at risk for severe malaria.
• Chloroquine or Mefloquine</td></tr>
<tr><td>Endemic areas</td><td colspan="2">Intermittent preventive treatment (IPTp) for all pregnant people
➢ SP: 4 doses ~1 month apart from 13 weeks (early 2nd trimester) until 38 weeks</td></tr>
<tr><td colspan="3">Treatment of pregnant people</td></tr>
<tr><td>Severe malaria</td><td colspan="2">➢ Artesunate IV or IM; switch to oral ACT when tolerated</td></tr>
<tr><td>Uncomplicated chloroquine-resistant malaria</td><td colspan="2">• Artemisinin-lumefantrine all trimesters or other ACTs if A-L is unavailable.
• 2nd/3rd trimesters: any ACT
• Quinine or clindamycin therapy are alternatives (all trimesters).</td></tr>
<tr><td colspan="3">Chemoprophylaxis for children at high risk of severe malaria (endemic areas): Intermittent curative doses of antimalarial therapy, regardless of whether the child is infected, to kill existing parasites and prevent new infection. WHO recommendation: SP plus amodiaquine (SP + AQ).</td></tr>
<tr><td colspan="2">Seasonal malaria chemoprevention (SMC)
Intermittent dosing during peak malaria transmission seasons</td><td>Perennial malaria chemoprevention (PMC)
Intermittent dosing at predefined intervals.</td></tr>
</table>

WHO malaria guidelines, 30 Nov 2024

Two WHO-approved Malaria vaccines: RTS,S/AS01 (Mosquirix) and R21/Matrix-M
Recombinant *Plasmodium falciparum*: circumsporozoite protein (CSP) central repeat region

The WHO recommends a 4-dose schedule for the prevention of *P. falciparum* malaria for children living in regions with moderate to high malaria transmission starting at 5 months of age:

3 doses at monthly intervals followed by a fourth dose to maintain efficacy

RTS,S/AS01: fourth dose 18 months after the third dose

R21/Matrix-M: fourth dose 12 months after the third dose

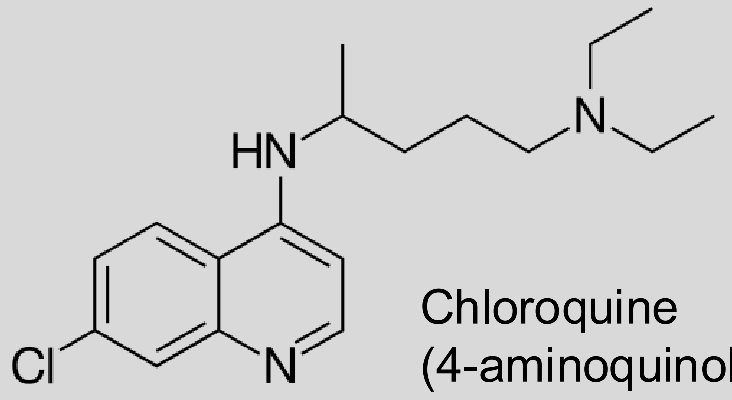
- Highly efficacious when given before high transmission season
- Safety demonstrated in clinical trials. Safety monitoring is continuing.
- Expected to have high public health impact when implemented broadly; 28 African countries plan to introduce WHO-recommended malaria vaccine as part of their national immunization programs.
- Cost effective (a few $US per dose), comparable to other malaria interventions and childhood vaccines.

Circumsporozoite protein is a multifunctional protein required for sporozoite development.

https://www.who.int/news/item/02-10-2023-who-recommends-r21-matrix-m-vaccine-for-malaria-prevention-in-updated-advice-on-immunization

✳Chloroquine Hydroxychloroquine

(weak bases)



Chloroquine
(4-aminoquinoline)

Blood schizonticide

Prophylaxis and treatment of chloroquine-sensitive malaria, erythrocytic stage.

May be used in pregnancy.

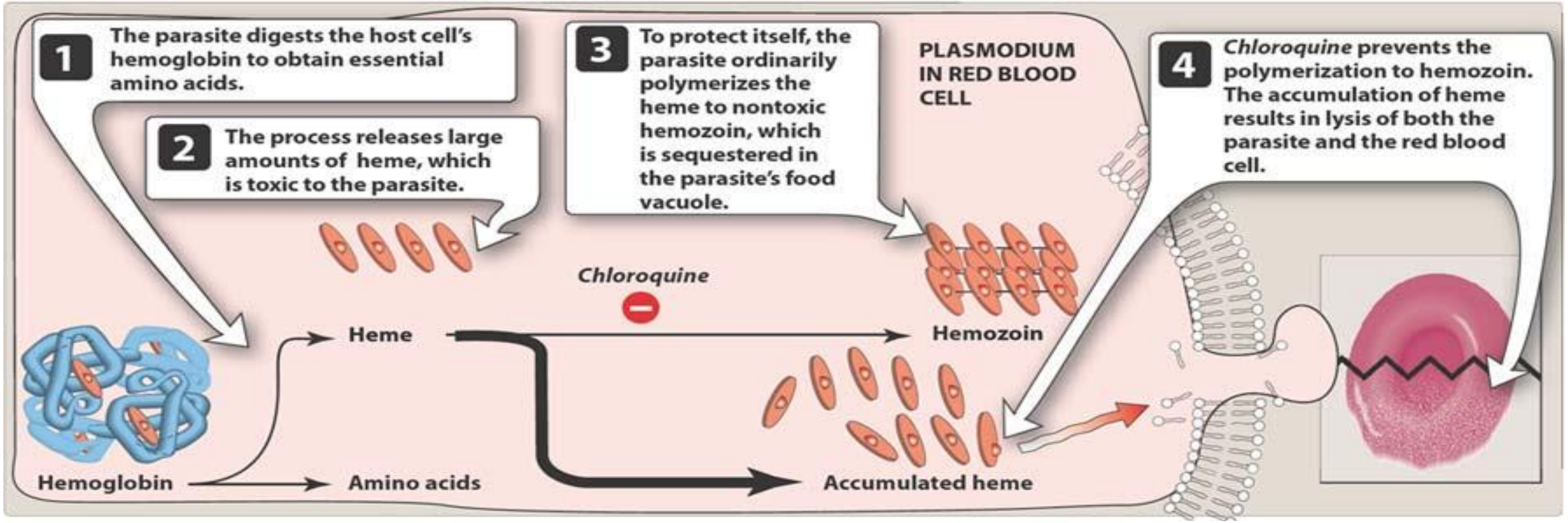
Worldwide chloroquine resistance.

Chloroquine-sensitive regions are Central America west of Panama Canal; the Dominican Republic; Haiti (resistance reported); and parts of the Middle East.

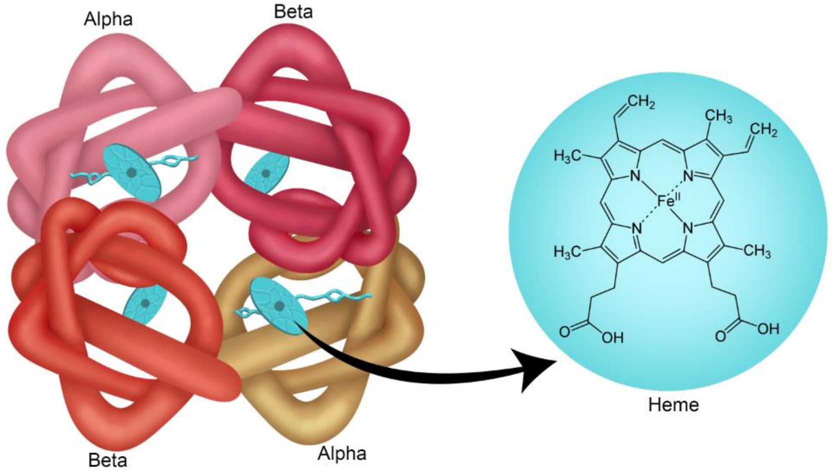
Chloroquine Pharmacology

PK	Oral (safest route) | very large Vd → loading dose (treatment) CYP3A4 metab. | urine, unchanged and metabs. Sequestered in liver, spleen, kidney, lung, brain, spinal cord, melanocytes $t½_{initial}$ 3-5 days; $t½_{terminal}$ 1-2 months	
	Prophylaxis 500 mg weekly x2 weeks prior to exposure Continue weekly and x4 weeks after exposure	Treatment: 1000 mg, then 500 mg 6, 24, and 48 hours after first dose Note: High dose – 2.5 g in 48 hours
MOA	Enters food vacuole of parasite → disrupts heme sequestration → oxidative damage to membrane and critical biomolecules (next slide)	
Resistance	1) *P. falciparum* chloroquine resistance transporter, PfCRT is expressed on the membrane of the food vacuole. **K76T point mutation in PfCRT gene** → ↓accumulation of drug in the parasites' food vacuoles ➢ K76T mutation is strongly associated with clinical treatment failure and is a molecular marker for surveillance of chloroquine resistance. 2) P-glycoprotein transporter (gene *pfmdr1*); *pfmrp* and other transporter gene mutations may also be involved in resistance to quinolines	

Possible action of chloroquine and hydroxychloroquine on the formation of hemozoin in *Plasmodium* species



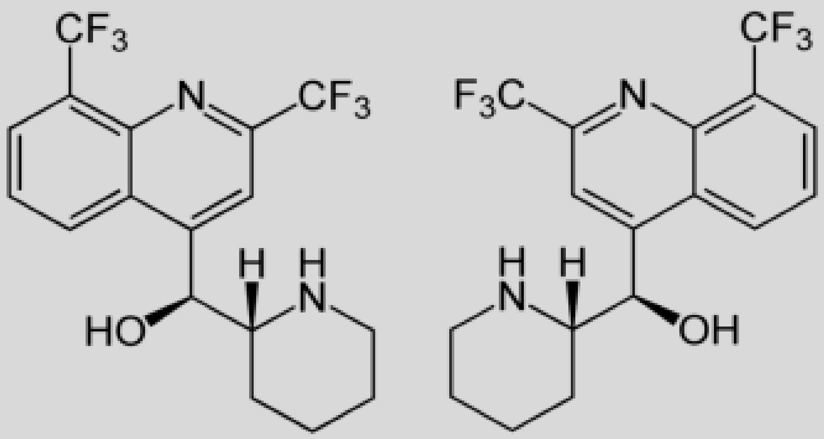
- Chloroquine and other quinolines concentrate in the highly acidic digestive vacuoles of susceptible *Plasmodium*
- Binds to heme → disrupt heme sequestration → failure to inactivate heme or enhanced toxicity of drug-heme complexes → oxidative damage to membranes, digestive proteases, or other critical biomolecules



Chloroquine Adverse Effects

Low prophylactic dose	Generally well tolerated at prophylactic doses GI irritation (take with food to decrease GI effects), rash, headaches	
High treatment dose	↑ risk of side effects Pruritis; diarrhea, nausea, vomiting; headache, visual disturbances; rash, photosensitivity	
Acute toxicity	Too rapid IV infusion: ***narrow safety margin*** (a single 30mg/kg or >5g dose usually fatal) The IV formulation is not available in the U.S.	
Long-term use (for treatment of autoimmune disorders)	***High cumulative doses*** Cardiovascular – arrhythmia, AV block, ECG changes; skin – blue-gray pigmentation, dermatitis, phototoxicity, rash, severe skin reactions; severe hypoglycemia; potentially irreversible ototoxicity, retinopathy, myopathy and peripheral neuropathy; agitation, anxiety, confusion, other neuropsychiatric effects	
Interactions (not a complete list)	Antacids, kaolin-pectin antidiarrheals → ↓ absorption	CYP3A4 inducers and inhibitors
Drug combos that could cause:	QT interval prolongation Concurrent use of drugs with the same toxicities increase the risk	

✳Mefloquine



four stereoisomers with similar antimalarial potency but different pharmacokinetic profiles

Blood schizonticide

Prophylaxis and treatment of chloroquine-resistant malaria, erythrocytic stage.

May be used in pregnancy.

Mefloquine PK, Actions, and Resistance

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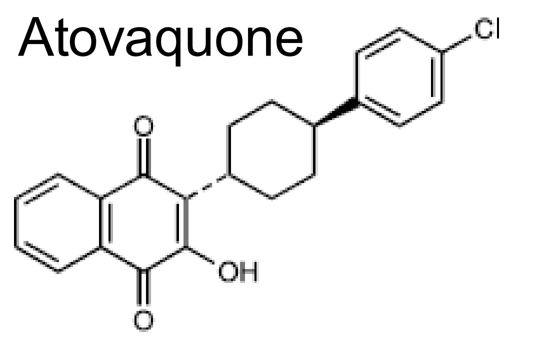
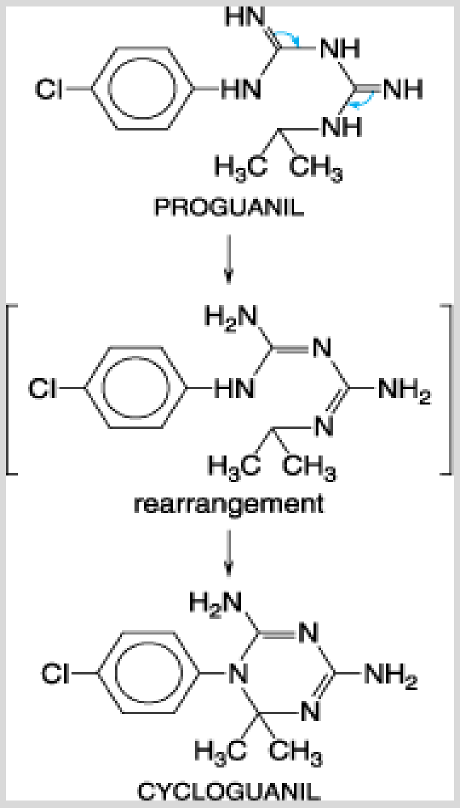
PK	Well absorbed, enhanced by food; widely distributed; CYP3A4, inactive metabolites excreted in bile/feces; $t_{1/2\,terminal}$ 2-4 weeks (adults)	
	Prevention: 250 mg once weekly	Treatment: 750 mg, then 500 mg 6 and 12 h later
MOA	unknown	
Resistance	Associated with amplification of *pfmdr1*	High level resistance in Mainland Southeast Asia

Neuropsychiatric effects: • Anxiety, depression, confusion may be a prodrome to more serious effects, such as paranoia, hallucinations, psychosis • Dizziness or vertigo, loss of balance can occur • Prophylactic use: If symptoms occur, discontinue mefloquine and use alternative. • Caution in activities requiring mental alertness (drivers, pilots, machinery operators, others) ***Mefloquine should not be prescribed for patients with major psychiatric disorders.***	• **Seizure disorder:** Mefloquine use increases risk of seizures. ***Mefloquine is contraindicated***. • Cardiac disease: Mefloquine causes ECG changes. Use with caution. • Heme: Agranulocytosis and aplastic anemia reported. • Liver: Mefloquine elimination may be prolonged in patients with hepatic disease. • Derm: Severe skin reactions • Hypersensitivity reactions And other adverse effects

Atovaquone





✳Atovaquone-✳Proguanil [Malarone]

Fixed-combination tablet

(atovaquone 250 mg, proguanil HCl 100 mg)

Blood schizonticide

Prophylaxis and treatment of chloroquine-resistant malaria, erythrocytic stage.

- Atovaquone and proguanil act synergistically.
- Atovaquone is highly active against erythrocytic- and primary hepatic-stage but not against hypnozoites of *P. vivax* and *P. ovale*.

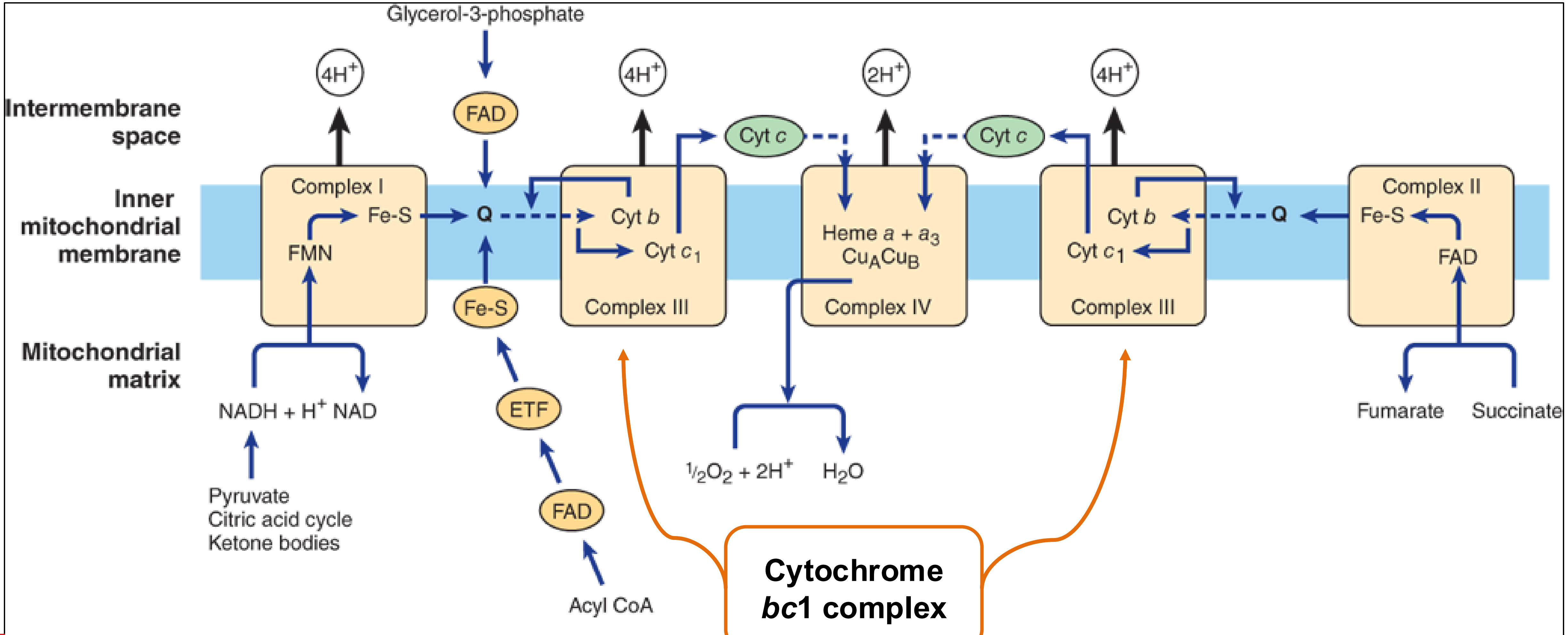
Not for use in pregnancy due to lack of safety data.

Proguanil is a biguanide that is converted to active cycloguanil.

Atovaquone-Proguanil Pharmacology

PK	**Atovaquone:** oral; >99% protein bound; not significantly metabolized; biliary/fecal excretion of unchanged drug; t½ 3 days	**Proguanil (prodrug):** oral; hepatic ***CYP2C19 to active cycloguanil***, urine excretion of proguanil and cycloguanil; t½ 12-20 hours
	Slow, erratic absorption and low bioavailability (take with fatty food)	
MOA	Collapse of the parasite mitochondrial membrane potential (next slide)	Cycloguanil inhibits the bifunctional dihydrofolate reductase–thymidylate synthetase
	Proguanil itself contributes to the collapse of the parasite mitochondrial membrane potential **Resistance:** Point mutations in cytochrome b gene within the cytochrome bc1 complex Resistance is not widespread.	
AEs	Generally well tolerated: Side effects include abdominal pain, nausea, vomiting (taking the drug with food may reduce GI effects); headache, dizziness; elevated serum transaminases (reversible), hepatitis reported	
DDIs	CYP2C19 inhibitors may reduce cycloguanil levels Other drug-drug interactions (mechanisms unknown); check literature prior to prescribing	

Atovaquone binds cytochrome bc_1 complex



Atovaquone binds cytochrome bc_1 complex on inner membrane of mitochondria

Inhibits electron transport → **collapses the parasite mitochondrial membrane potential**.

Cycloguanil (active metabolite of proguanil) selectively inhibits the bifunctional **dihydrofolate reductase–thymidylate synthetase** of sensitive plasmodia → inhibits nucleotide, DNA and RNA synthesis

Proguanil (parent drug) accentuates mitochondrial membrane-potential-collapsing action of atovaquone against *P. falciparum* but it displays no such activity by itself.

✳Sulfadoxine and ✳Pyrimethamine (SP)

Available only in a fixed-dose tablet [Fansidar]

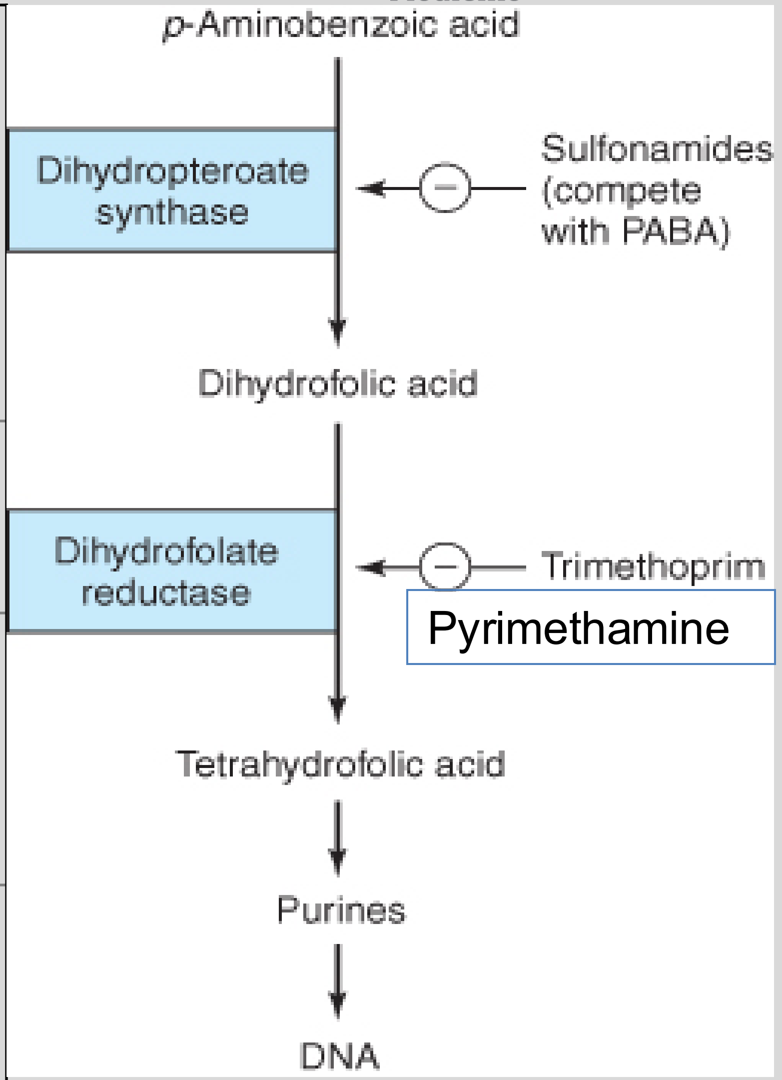
- No longer recommended for travelers for prophylaxis or treatment of *P. falciparum* malaria due to widespread resistance.
- The product has been discontinued in the U.S.

Blood schizonticide

- Sulfadoxine and pyrimethamine act synergistically to inhibit folate synthesis.
- The long half-lives of the drugs prolongs exposure, which allows selection of resistant organisms. Additionally, repeated exposure to SP in malaria endemic regions contributes to resistance.
- Avoid in pregnancy first trimester due to pryimethamine adverse events observed in animal reproduction studies.
- WHO recommends SP for intermittent preventive treatment in pregnancy for pregnant patients without HIV who live in areas of high malaria transmission starting at 13 weeks .

Sulfadoxine and Pyrimethamine Pharmacology

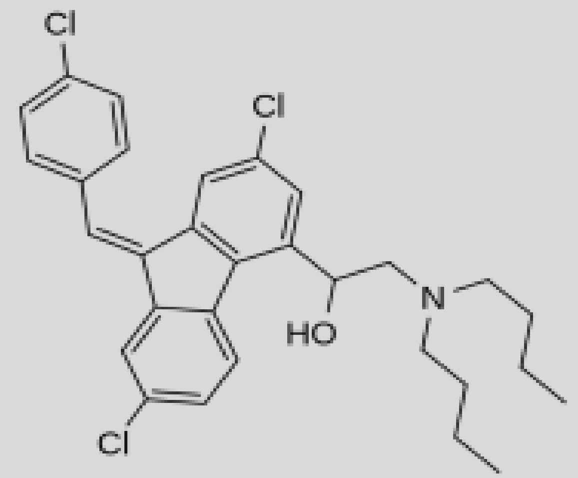
PK	**Sulfadoxine:** oral, slowly absorbed; ~90% protein bound; metabolism by N-acetylation; urine >90% unchanged drug; t½ 8 days	**Pyrimethamine:** oral; well absorbed; ~90% protein bound; hepatic metabolism; urine, ~25% unchanged drug; t½ ~4 days
MOA	Sulfadoxine inhibits dihydropteroate synthase (DHPS)	Pyrimethamine inhibits dihydrofolate reductase (DHFR)
	Resistance: Point mutations in Pf-DHPS gene reduces drug binding.	**Resistance:** Point mutations in Pf-DHFR genes reduces drug binding.
AEs	Potentially severe skin rashes: erythema multiforme, Stevens-Johnson syndrome, toxic epidermal necrolysis, drug reaction with eosinophilia and systemic symptoms (DRESS) (sulfonamides class effect)	High doses can cause megaloblastic anemia, leukopenia, thrombocytopenia. Caution in patients with folate deficiency, hepatic or renal impairment, history of seizure disorder. Probably safe in G6PD deficiency at therapeutic doses.



Katzung Basic & Clinical Pharmacology 14e, 2018 Figure 46-2

✳Lumefantrine

Only used in fixed-combination with artemether [Coartem]



Lumefantrine Pharmacology

PK	Oral with high fat meal (16-fold increase in absorption) Highly protein bound; CYP3A4, inactive metabolite; t½ 3-6 days
MOA	Unknown. Laboratory assay data suggests it decreases hemazoin formation.
Activity	Blood schizonticide
Use	**Fixed combination tablet: artemether and lumefantrine = Coartem** **Treatment of uncomplicated malaria; 3-day course**
AE	Well tolerated; mild QT interval prolongation
	Cautions – ECG monitoring may be advised in the following conditions: • Patients with long QT syndrome or symptomatic arrhythmias • Administered with other drugs that can prolong the QT interval • Concomitant use of CYP3A4 inhibitors • Concomitant use of CYP3A4 inducers is contraindicated → ↓ efficacy

Short-acting blood schizonticides for the treatment of *P. falciparum* and non-falciparum malaria

Short-acting antimalarials are *always* used with a long-acting antimalarial drug.

- ✳Artemisinin derivatives
- ✳Quinine

✳Artemisinin derivatives

- ✳Artemether
- ✳Artesunate
- Artemisinin
- Dihydroartemisinin

Artemisinin combination therapy (ACT):

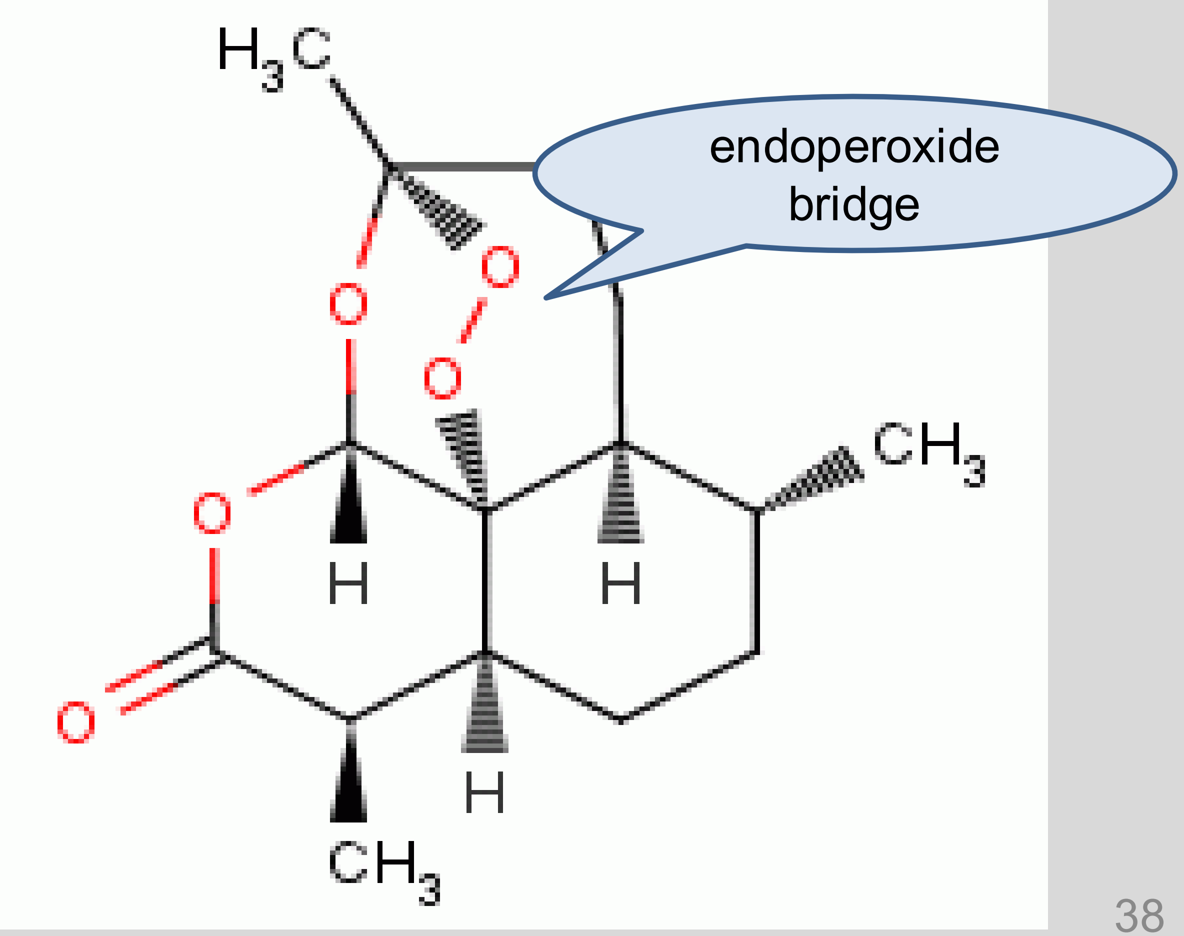
Several fixed-dose products of artemisinin with a long-acting antimalarial are available.

Products available in the US:

✳Artemether-lumefantrine [Coartem]

✳Artesunate

Artemisia annua - Qinghao



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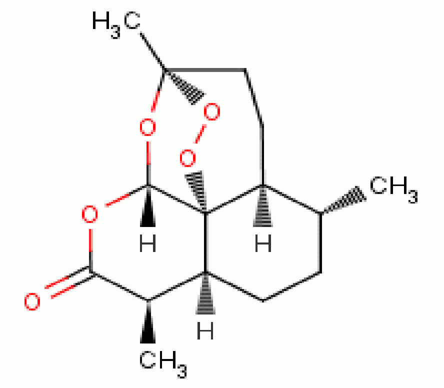
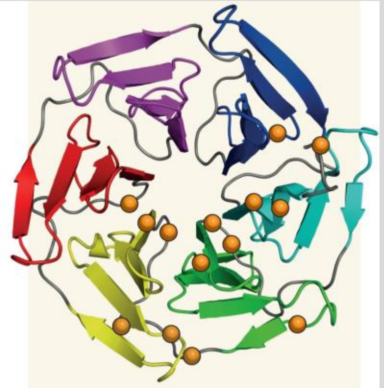
Artemisinins Class Pharmacokinetics (US products)

	Artemether	Artesunate (prodrug)
A	Oral with high fat meal (2-3-fold increase in absorption)	IM, IV, (rectal in some countries) prodrug of DHA
D	Highly protein bound Widely distributed	Highly protein bound Widely distributed
M	Hepatic metabolism (CYP3A4 mainly) to dihydroartemisinin (DHA) active metabolite	Artesunate → plasma esterases → DHA • DHA → glucuronide metabolite, inactive
E	Urine as inactive metabolites	Urine, DHA glucuronide metabolite
t½	~2 h | Tmax ~2 h	DHA 1.3 h
Recall:	**Coartem:** **Artemether is combined with lumefantrine.**	**Artesunate is used for the treatment of severe malaria.**

Artemisinins Activity





Proposed mechanism of action:

1. Artemisinin binds to free heme in the food vacuole released after digestion of hemoglobin by the parasite.
2. The drug interacts with the heme iron forming covalent complexes, possibly disrupting heme detoxification and increasing oxidative stress.
3. Oxidative stress causes structural damage and inhibition of nucleic acid and protein synthesis.
4. ***The endoperoxide bridge is essential for activity***.

Activity

- Active against asexual erythrocytic stages of chloroquine-sensitive and chloroquine-resistance, all plasmodia spp
- Gametocidal activity
- No activity against primary or latent liver stages
- Nonplasmodial activity:

Antiparasitic activity demonstrated against *Leishmania major*, *Toxoplasma gondii* and *Schistosoma* species

Figure right: Propeller mutations in propeller blades of K13 Kelch protein (orange spheres)
https://www.nature.com/articles/nature12845

Artemisinin partial resistance to *P. falciparum* is emerging in Greater Mekong Subregion, Papua New Guinea, South America, and Eastern Africa.

Mechanism undefined

Possible mechanism

- Point mutations in the *PFkelch* BTB/POZ propeller domain (*PFk13*) are associated with delayed parasite clearance.
- Down-regulation of metabolic activity in ring stages →…→ reduced activation and conversion of artemisinin drugs to reactive intermediates
- Therapy may be extended from 3 days to 6 days and follow with daily smears until negative.

ACT Treatment of Malaria

WHO recommendations:

- Artemisinins should be used ***only in combination with longer-acting antimalarials*** to deter emergence of resistance and prevent recrudescence.
- ACT: ***First-line*** treatment of severe malaria and uncomplicated malaria in all populations

Uncomplicated malaria	Artemether-Lumefantrine 80 mg/480 mg tabs (4 tablets) 2x daily for 3 days Weight-based dosing for infants and children First 2 doses 8 hours apart, ideally
Severe malaria	Artesunate I.V. bolus or I.M. x 4 doses (0,12, 24, and 48 hours) then 1x daily – Treat until parasitemia <1%; complete course orally when tolerated by patient – Oral therapy must begin ≤4 hours after the last dose of I.V. artesunate.
Reduce gametocyte transmission	In low transmission regions, a single dose of primaquine on Day 1 of treatment along with ACT (except pregnant people and breastfeeding infants <6 months old)

WHO Guidelines for the Treatment of Malaria, 30 November 2024
https://www.who.int/teams/global-malaria-programme/guidelines-for-malaria

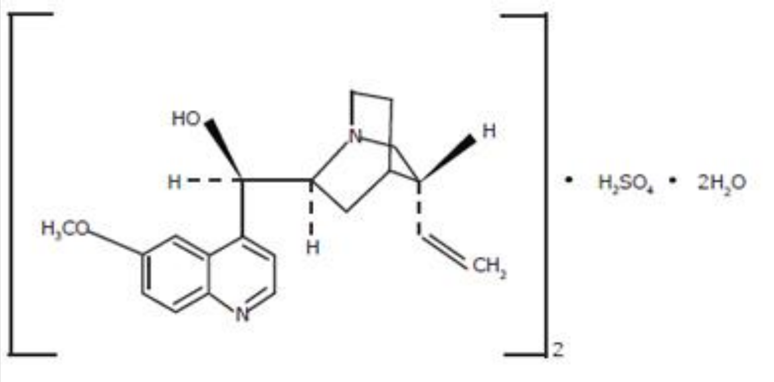
Artemisinins Side Effects

- Generally well tolerated
- Most common: nausea, vomiting, diarrhea, and dizziness
- Hematologic: Dose-related reversible neutropenia, anemia, and acute hemolysis
- Hepatic: transient increases in serum transaminases
- Blackwater fever:

  Artesunate I.V. in the treatment of severe malaria

  - Bursting red blood cells release hemoglobin into the bloodstream and urine, which turns dark red or black. Kidney failure can result.
  - Delayed-onset (may occur 1-3 weeks after therapy)
  - May be severe (reported in up to 10% of travelers)
  - Higher likelihood with higher parasite density

✳Quinine sulfate



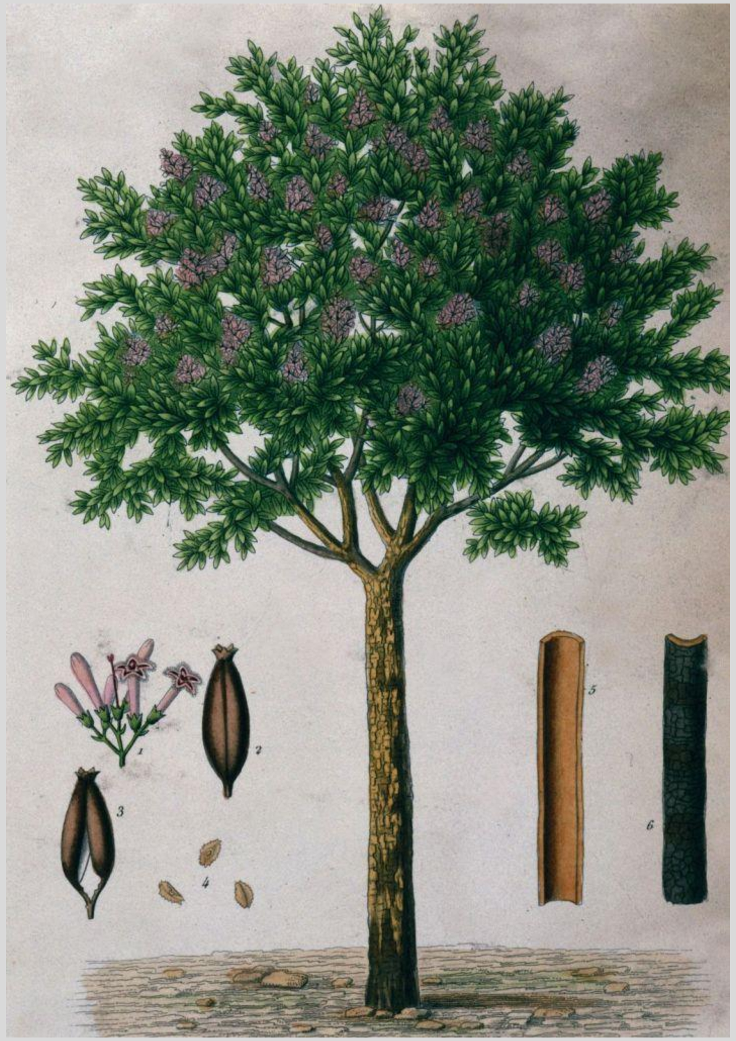
A weak base

Blood schizonticide

Mechanism is unknown.

Short-acting drug for the treatment of chloroquine-resistant malaria, erythrocytic stage.

May be used in pregnancy but not preferred. Quinine can cause recurrent hypoglycemia. Pregnant patients are especially susceptible to hypoglycemia.



Quinine is obtained from the bark of the cinchona tree. (native to South America)

Quinidine is more potent stereoisomer of quinine developed for the treatment of arrhythmias.
Quinidine I.V. was used in the past for the treatment of severe malaria.
Quinidine is cardiotoxic and manufacture of the IV form of the drug has been discontinued.

Quinine Pharmacokinetics and Therapeutic Uses

A	Oral; readily absorbed | Tmax 3-8 (IV formulation not available in US)	Therapeutic use: uncomplicated *P. falciparum*-resistant malaria • an alternative to ACT • Quinine is also effective for non-falciparum malaria. • Coadministered with long-acting antimalarial agent: Doxycycline or Tetracycline or Clindamycin (young children / pregnancy) Dosing, adults: (CDC recommendations) • 2 caps (648 mg) 3x daily for 3 days • or 7 days if infection was acquired in Southeast Asia FYI: Babesiosis, a protozoan parasite transmitted by deer ticks: Treatment 7 to 10 days • quinine and clindamycin (alternative) • atovaquone and azithromycin (preferred)
D	• α1-acid glycoprotein binding ~70-95% ➢ increased α1-acid glycoprotein levels in severe disease P-glycoprotein inhibitor	
M	CYP3A4 mainly; forms a less active metabolite Bioavailability ~85%	
E	Urine ~20% unchanged drug • Severe long-term renal impairment: adjust dosage	
t½	Healthy adults ~11h; ~18 h severe malaria • Shorter in healthy children • ~18 h in elderly individual	

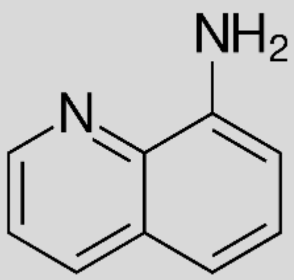
Quinine Toxicities

Cinchonism (from cinchona)	Mild cinchonism: occurs with therapeutic doses • Tinnitus, headache, vasodilation, sweating, nausea, dizziness; visual disturbance Severe cinchonism: marked effects after prolonged therapy (quinine should be discontinued)
Hypoglycemia	Can occur at therapeutic doses. *Hypoglycemia is especially dangerous in patients with* ***severe malaria and pregnant patients****, who have increased sensitivity to insulin.* Combined effects of: — quinine, increased consumption of glucose by host and parasites, failure of hepatic gluconeogenesis —
Hypotension	Severe with rapid IV infusion of quinine or quinidine
Blackwater fever	Malarial hemolytic anemia with quinine treatment • Massive intravascular hemolysis, hemoglobinemia, hemoglobinuria, anuria, *renal failure, death*
Thrombocytopenia	Drug-induced immune-mediated reaction • Life-threatening hemolytic uremic syndrome / thrombotic thrombocytopenic purpura (HUS/TTP)
Hypersensitivity reactions	Skin rashes, urticaria, angioedema
IV Administration	Quinine, like quinidine, is cardiotoxic when administered IV. QTc interval prolongation→↑ risk of ventricular arrhythmia (torsades de pointes)
Pregnancy	Quinine appears to be safe in pregnancy ➢ Glucose levels must be monitored because of the increased risk of hypoglycemia.

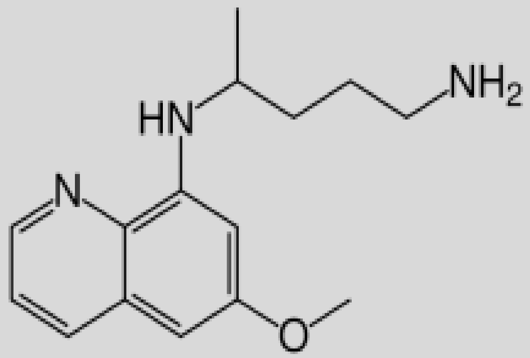
Potentially serious quinine-related drug-drug interactions

Drugs with similar toxicities	• QT interval prolongation: lumefantrine, antiarrhythmics, macrolide antibiotics, azole antifungal agents, and many more
Metabolism-related Transport-related	CYP3A4 inducers and inhibitors can affect quinine levels. P-glycoprotein inhibition by quinine can affect levels of P-gp substrates. • digoxin, cyclosporine, many cancer drugs, and many many more
Warfarin	Quinine may enhance the anticoagulant effects
Neuromuscular blockers	Enhance neuromuscular block (avoid combination) The mechanism is not understood.
Antacids	May decrease serum concentration of quinine. Avoid
and many more	

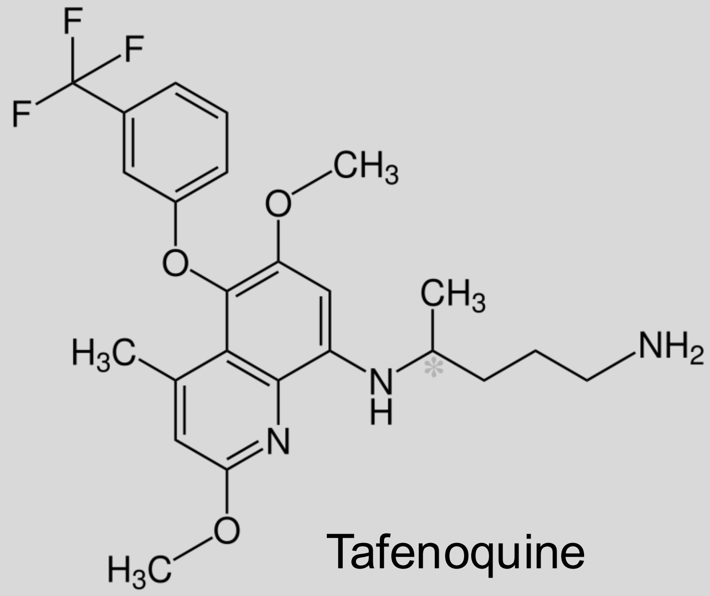
Tissue schizonticides
for the prevention and treatment of latent liver stages
✳Primaquine
✳Tafenoquine



8-aminoquinoline



Primaquine



Tafenoquine

Primaquine and Tafenoquine Pharmacokinetics

	Primaquine	Tafenoquine
A	Oral, well absorbed T_{max} 1-3 hours	Oral, absorption increased with high fat meal T_{max} 12-15 hours
D	Widely distributed to tissues	Protein binding >99% V_d 1600-2470 L
M	Hepatic, CYP1A2 to active metabolite Carboxyprimaquine also a substrate of CYP2D6, 3A4	Not available (not a substrate of CYP450 enzymes or phase 2 enzymes)
E	Urine, metabolites	Not available
t½	7 hours (range ~4-10 hours)	**15 days**

Mechanism of action: Unknown

Primaquine and Tafenoquine Actions and Uses

G6PD testing prior to therapy is required. Dosing recommendations are for adults with normal G6PD levels

Liver stages: Active against the hepatic stages of all human malaria parasites ***including the dormant hypnozoite stages of P. vivax and P. ovale***	**Gametocidal** In low transmission areas, ➢ A single dose of primaquine following ACT reduces *P. falciparum* gametocyte carriage and transmission and is believed to be safe in people with G6PD deficiency.

Standard therapy (radical cure) of *P. vivax* and *P. ovale* malaria
- Chloroquine (or hydroxychloroquine) is administered to eradicate erythrocytic forms
  - Primaquine has only modest activity against asexual erythrocytic stages.
  - Tafenoquine has substantial activity against asexual erythrocytic stages but is not indicated for acute infection.
- Primaquine (infants, children, adolescents, adults): 30 mg once daily for 14 days
- Tafenoquine (≥16 years of age): 200 mg single dose on day 1 or 2 of chloroquine therapy

Terminal prophylaxis after departure from malarious region to reduce the likelihood of relapse:
- Primaquine: 30 mg once daily for 14 days
- Tafenoquine: 200 mg single dose 7 days after completion of maintenance regimen

Chemoprophylaxis while in endemic areas
- Primaquine: Once daily beginning 1-2 days before travel and continuing for 7 days after return
- Tafenoquine: Loading dose followed by once weekly for up to 6 months (people ≥18 years of age)

Primaquine and Tafenoquine hemolytic anemia related to G6PD deficiency

G6PD deficiency →

Hemolytic anemia and methemoglobinemia (NADPH deficiency)

- >200 million people worldwide, ~11% of African Americans

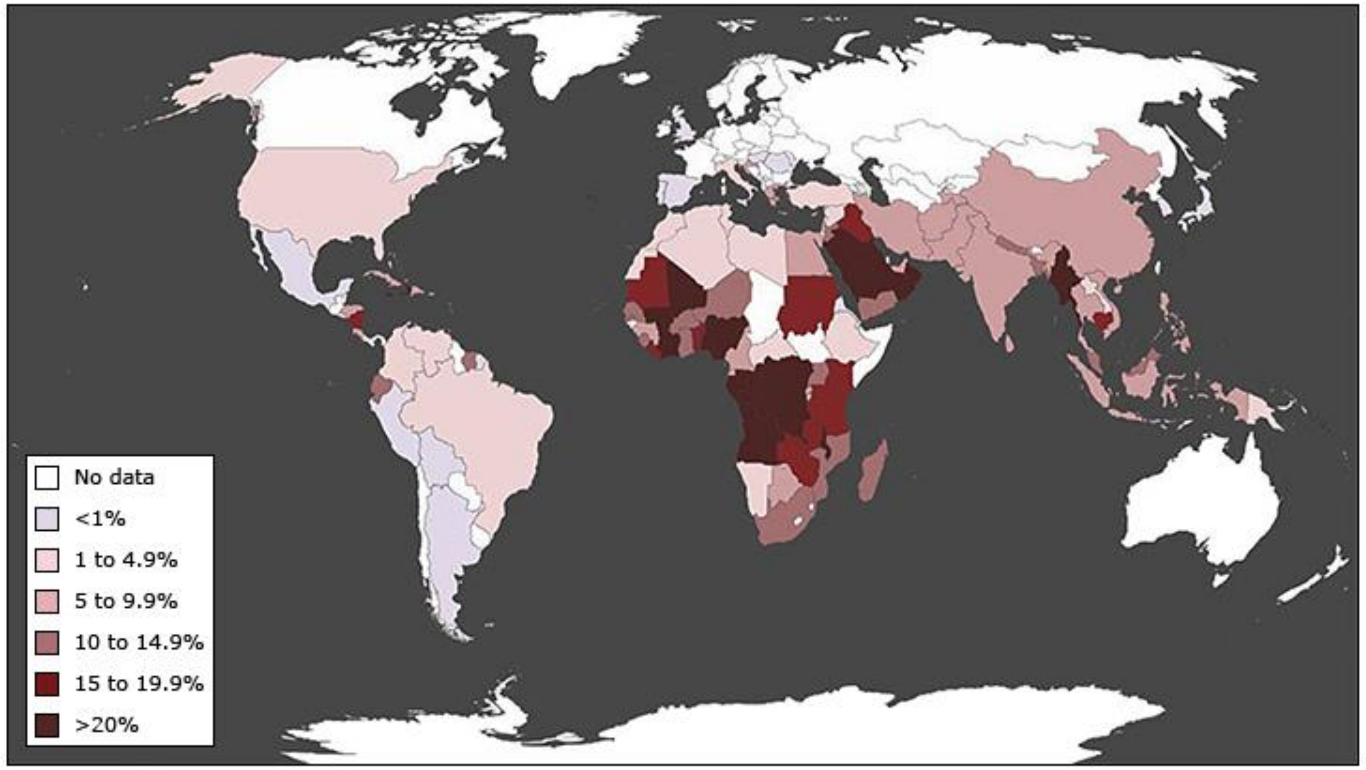
Test for G6PD deficiency before PQ or TQ administration:

- PQ: Full dose for people with G6PD activity ≥70%; modified dosing for people with 30-70% G6PD activity
- TQ: Use only in patients with ≥70% G6PD activity.

Contraindications:

- Acutely ill patients predisposed to granulocytopenia (RA, SLE)
- Other potentially hemolytic drugs
- **PREGNANCY** Risk of fetal hemolysis if the fetus has G6PD deficiency.

Map showing the prevalence of G6PD deficiency throughout the world

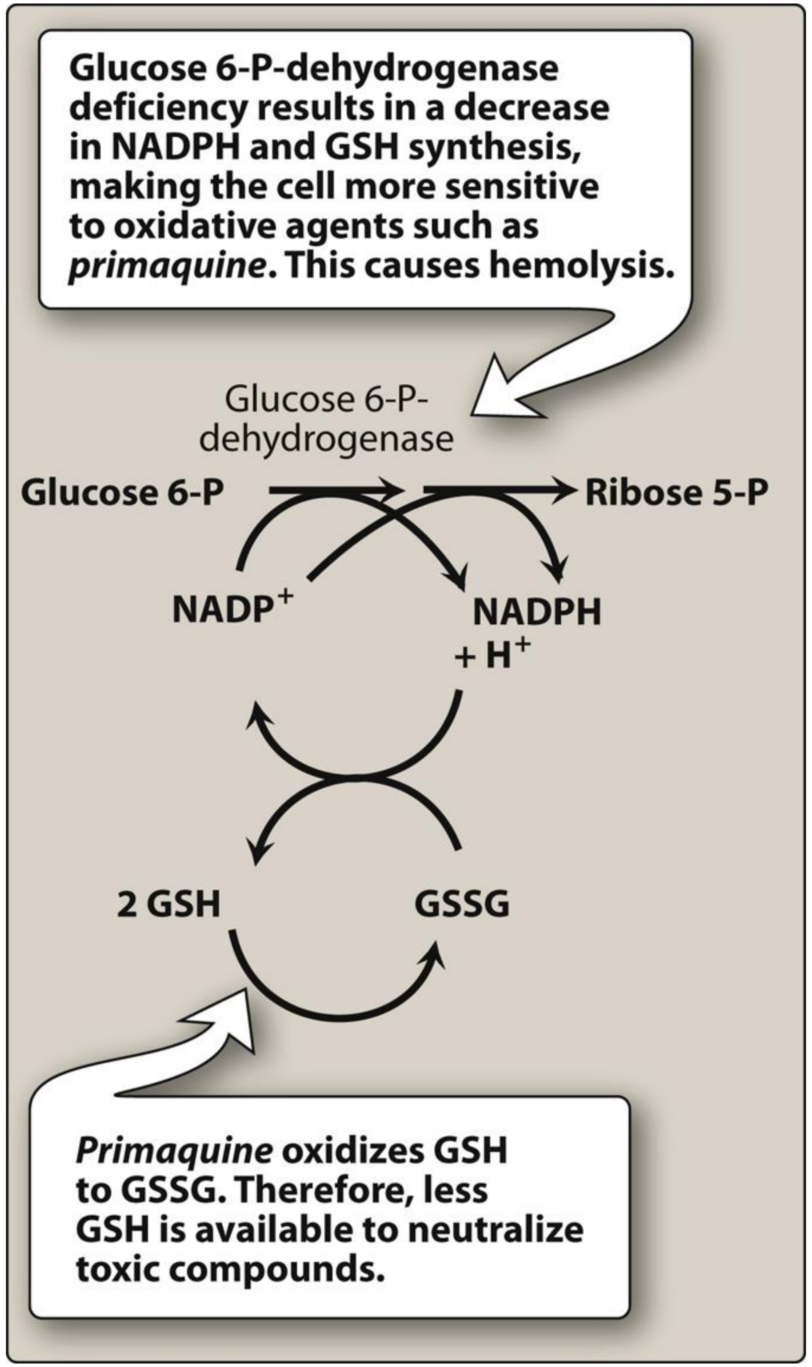


The map shows the prevalence of G6PD deficiency variants throughout the world, with darker colors indicating higher prevalences. It was constructed from several sources; see the source document for details. Refer to UpToDate for additional information on the epidemiology, clinical presentation, and diagnostic evaluation for G6PD deficiency.

G6PD: glucose-6-phosphate dehydrogenase deficiency.

UpToDate®

Mechanism



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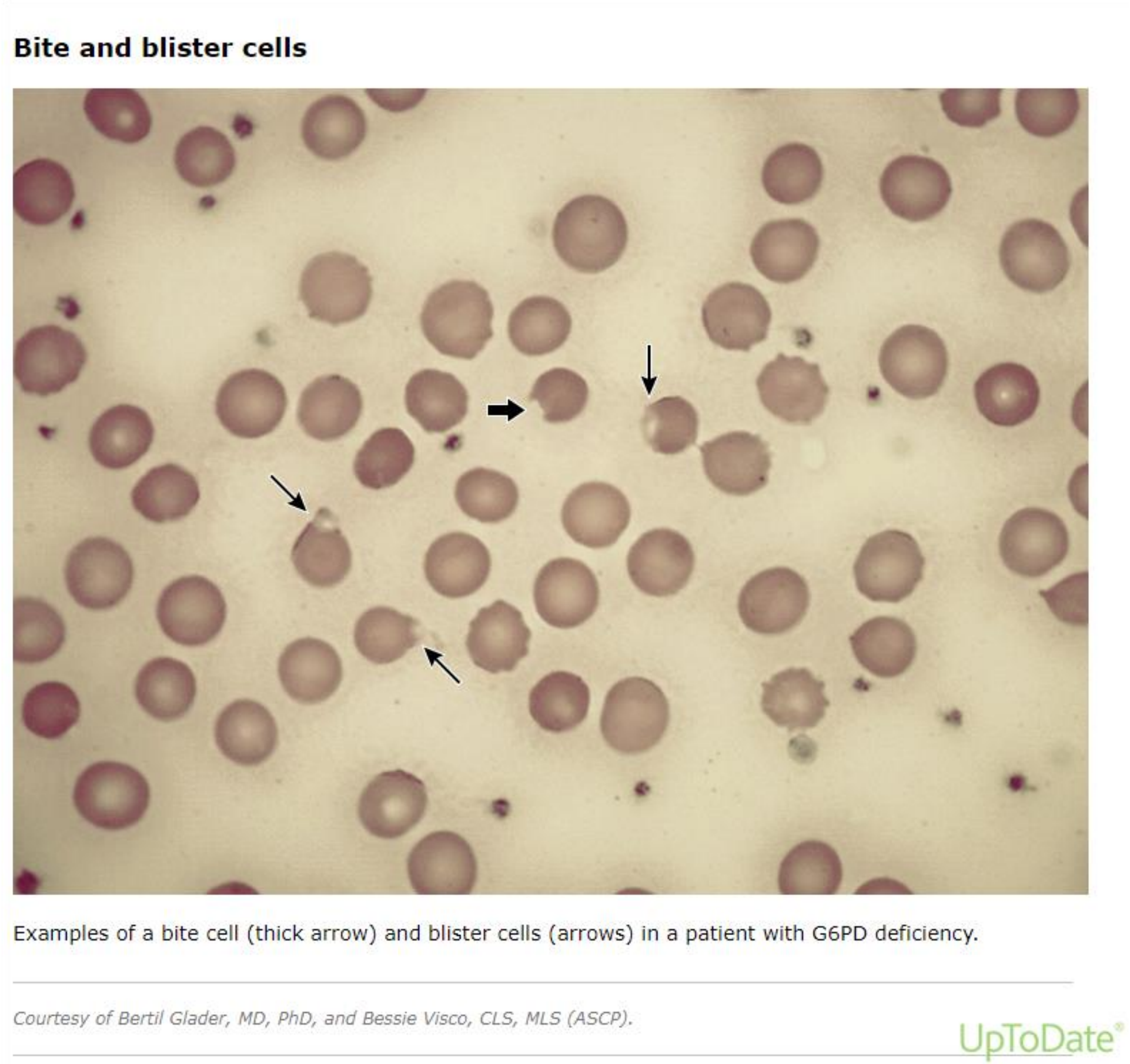
Lippincott® Illustrated Reviews: Pharmacology, 8e, 2023 Figure 35.6

Hemolysis



Inherited disorder

Genetic defect in the red blood cell enzyme G6PD

G6PD generates NADPH and protects RBCs from oxidative injury

Oxidant injury → hemolysis

Severity determined by the drug, the degree of enzyme deficiency, and other factors (*eg* infection)

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Antibiotics with antimalarial activity

✳**Doxycycline**

Tetracycline

✳**Clindamycin**

Doxycycline and Clindamycin Actions

Antimalarial activity:

- Blood stage: Slow-acting schizonticides
- Liver stage: Do NOT eliminate liver stage infection

Mechanism: *Delayed death mechanism*

Specifically impair expression of the apicoplast gene → abnormal cell division → ***death of the progeny*** of the drug-treated *Plasmodium* parasites.

Thus, onset of antimalarial activity is delayed.

Dahl EL, et al, 2006; Ramya TNC, et al, 2007; Dahl EJ, et al, 2007

Doxycycline or tetracycline:

- Prophylaxis: Short-term prophylaxis in chloroquine-resistant areas
- Treatment: Adjunct to quinine or artesunate for uncomplicated or severe malaria

Clindamycin: treatment alternative to doxycycline in children <8 y.o. and pregnant people

Doxycycline, tetracycline, and clindamycin are not effective as single agents in the treatment of acute malaria due to their slow action.

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Doxycycline and Clindamycin Common Adverse Effects

Doxycycline and tetracycline:

GI upset; yeast infections; photosensitivity

Clindamycin:

Diarrhea and skin rash, particularly in HIV-infected patients

Pseudomembranous colitis caused by a toxin released from *Clostridium difficile* can occur

Tetracyclines are contraindicated in children <8 years old and pregnant people.

Tetracyclines cross the placenta and accumulate in long tubular bones and developing teeth, causing permanent discoloration.

FYI: Doxycycline has not been found to cause tooth staining in pediatric patients treated for rickettsial disease when used for ≤21 days. Doxycycline is the most effective treatment for rickettsial disease.

Check your knowledge

1. Suzie is a 22-year-old healthy woman planning a 2-week mission trip to West Africa with her church group. NKDA. She is provided with medication to prevent malaria and directed to start taking it once daily with a fatty food two days before traveling, continue for the duration of her stay, and for seven days after her return. She packs lightweight long-sleeved shirts and long pants, DEET-based bug spray, and insecticide treated bed nets. What is the mechanism of the drug that was prescribed? Why should the drug be taken with a fatty food?

Check your knowledge

2. Andy is a 50-year-old male being sent by his company to Central Africa to oversee a new division. He has a history of major depression managed with meditation, exercise, diet, and country line dancing. NKDA. Which antimalarial prophylaxis medication should be avoided for Andy and why?

3. What antimalarial drugs would be appropriate for Andy?

Check your knowledge

4. JP is 38 years old and planning a 10-day trip to Asia where the predominant malarial parasite is *P. vivax*. JP is healthy with no medical conditions. NKDA. A prescription is provided for an antimalarial drug for prophylaxis of parasitic infection of the red cells. What is the drug? What are the directions for taking it? Why?

5. When JP returns to the U.S., he is provided a prescription for a long-acting drug taken as a single dose and directed to continue the antimalarial drug he was taking during his stay once weekly for an additional four weeks. What is the purpose of the additional drug? What are two contraindications to its use?

Check your knowledge

6. AK is a 25-year-old woman invited to a family wedding in West Bengal in one month. She was born in the U.S. and is pregnant at 23 weeks gestation (G1,P0). She states that she is excited to see her family, whom she has not visited in ten years. For malaria prophylaxis, which of the following is the best option for her?

   A. Atovaquone-proguanil
   B. Chloroquine
   C. Mefloquine
   D. Doxycycline
   E. Do not travel to South Asia

Check your knowledge

7. Angelica is a previously healthy 4-year-old female who presents to the ED with fever and chills, rapid heart rate, rapid breathing rate, headache, fatigue, nausea, diarrhea, and joint and muscle pains for 24 hours. Signs and symptoms of severe disease are absent. History is significant for recent travel to Haiti. Physical exam and laboratory findings are consistent with *P. falciparum* infection. The patient is able to take oral medication. Which of the following is the first-line therapy for this patient?

   A. Artemisinin-lumefantrine
   B. Artesunate
   C. Chloroquine
   D. Quinine-doxycycline
   E. Primaquine

Summary

Generally Accepted Mechanisms of Action of the Commonly Used Antimalarial Agents

Chloroquine Hydroxychloroquine	Drug concentrates in the highly acidic digestive vacuoles of susceptible *Plasmodium*, where it binds to heme and disrupts its sequestration as hemozoin
Mefloquine	Unknown
Atovaquone	Selective action on the mitochondrial cytochrome bc_1 complex to inhibit electron transport and collapse the mitochondrial membrane potential
Proguanil	1. Active metabolite selectively inhibits the bifunctional dihydrofolate reductase–thymidylate synthetase 2. Parent compound accentuates the mitochondrial membrane-potential-collapsing action of companion drug (proguanil)
Sulfadoxine	Competitively inhibits PABA at *Plasmodium* dihydropteroate synthase
Pyrimethamine	Potently inhibits *Plasmodium* dihydrofolate reductase
Lumefantrine	Unknown
Artemisinin derivatives	Artemisinins interact with the heme iron after hemoglobin digestion by parasite. Covalent complexes form, possibly disrupting heme detoxification and increasing oxidative stress, which causes structural damage and inhibition of nucleic acid and protein synthesis. The endoperoxide bridge is essential for activity.
Quinine	Unknown
Primaquine Tafenoquine	Unknown

CHEMOPROPHYLAXIS OF MALARIA

Drug therapy	Chemoprophylactic use	Comments
Chloroquine	Use only in limited regions of chloroquine-sensitive *P. falciparum*, and for prophylaxis of *P. vivax*, *P. ovale*, and *P. malariae*.	Taken once weekly; can be used in all trimesters of pregnancy
Mefloquine	For prophylaxis of chloroquine-resistant, mefloquine-sensitive *P. falciparum* Once weekly dose Start 1-2 weeks before travelling to endemic region and continue for 4 weeks after return	May be used in pregnancy
Mefloquine is not a good choice for individuals with major psychiatric conditions, seizure disorder, or cardiac conduction abnormalities.		
Atovaquone-proguanil	For chloroquine-resistant *P. falciparum*; start 1-2 days before entering an area where malaria transmission occurs and continue for 7 days after leaving the region; taken daily	Very well tolerated Avoid in pregnancy and children <5kg Costly
Doxycycline Clindamycin: young children, pregnancy	Prophylaxis of chloroquine-resistant *P. falciparum*; take 1 day before entering endemic area and continue for 4 weeks after leaving the region; taken daily	The least expensive antimalarial Avoid in pregnancy and children <8 years old Doxycylcine causes photosensitivity and GI distress
Sulfodoxine-Pyrimethamine	SP recommended for intermittent preventive treatment (IPT) in pregnancy after 1st trimester: 4 doses ~28 days apart from 2nd trimester until 38 weeks	For infants and children at high risk of severe malaria in malaria-endemic areas, seasonal and perennial malaria chemoprevention (SMC / PMC) with SP plus amodiaquine
Primaquine Tafenoquine	For prophylaxis of *P. vivax* in travelers going to regions with >90% *P. vivax*; PQ: Once daily dose; Start 1-2 days prior to entering endemic area and continue for 7 days after return TQ: age ≥18yo; Loading dose 3 days before exposure, then once weekly starting 7 days after the loading dose.	G6PD deficiency testing required prior to use. Contraindicated in pregnancy and breastfeeding unless the infant has tested negative for G6PD deficiency May cause GI distress.

Malaria Treatment Recommendations: 3-day regimens usually

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Chloroquine or hydroxychloroquine	48-hour treatment regimen for uncomplicated P*. falciparum* malaria **in chloroquine-sensitive areas only** and for treatment of non-falciparum malaria by *P. vivax / P. ovale / P. malariae /* and *P. knowlesi*.
Artemisins	Uncomplicated malaria: oral artemisinin combination with a long-acting agent Artemether-lumefantrine in the US; lumfantrine t½ 72-144 hours; can cause mild QT prolongation Treatment of severe malaria: artesunate IV/IM; switch to oral combination when appropriate
Mefloquine	Available outside U.S as fixed-combination with atesunate or artemether for treatment of uncomplicated malaria (2-day treatment regimen)
Atovaquone-proguanil	Oral option for uncomplicated malaria and after treatment of severe malaria with artesunate.
Quinine	Quinine plus doxycycline (or clindamycin) alternative treatment of uncomplicated malaria Quinine IV is available outside the U.S. for severe malaria with artesunate.
Primaquine Tafenoquine	Radical cure: Eradication of *P. vivax* and *P. ovale* hypnozoites in combination with schizonticide (Primaquine daily x14 days; Tafenoquine single dose (long half-life) Terminal prophylaxis: After departure from malaria region (PQ daily x14 days; TQ single dose) Testing for G6PD deficiency prior to use is required.

Village Life

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Check your knowledge

1. Suzie is a 22-year-old healthy woman planning a 2-week mission trip to West Africa with her church group. NKDA. She is provided with medication to prevent malaria and directed to start taking it with a fatty food once daily two days before traveling, continue for the duration of her stay, and for seven days after her return. She packs lightweight long-sleeved shirts and long pants, DEET-based bug spray, and insecticide treated bed nets. What is the mechanism of the drug that was prescribed? Why should the drug be taken with a fatty food?

- Atovaquone-proguanil
- **Atovaquone** binds cytochrome bc_1 complex on inner membrane of mitochondria, which inhibits electron transport, which leads to the **collapse of the parasite mitochondrial membrane potential**.
- **Cycloguanil** (active metabolite of proguanil) selectively inhibits the bifunctional **dihydrofolate reductase–thymidylate synthetase** of sensitive plasmodia.
- **Proguanil** (parent drug) **accentuates mitochondrial membrane-potential-collapsing action** of atovaquone against *P. falciparum* but it displays no such activity by itself.
- Atovaquone-proguanil should be taken with a fatty food to improve absorption and may help relieve the GI side effects – abdominal pain, nausea, vomiting.

Check your knowledge

2. Andy is a 50-year-old male being sent by his company to Central Africa to oversee a new division. He has a history of major depression managed with meditation, exercise, diet, and country line dancing. NKDA. Which antimalarial prophylaxis medication should be avoided for Andy and why?

- Chloroquine: Sub-Saharan Africa is a region with widespread chloroquine-resistant *P. falciparum.*
- Mefloquine: Neuropsychiatric effects – anxiety, depression, confusion, which may be a prodrome to more serious effects, such as paranoia, hallucinations, psychosis. Mefloquine is not recommended for patients with a history of major psychiatric disorders.
- Mefloquine can also cause dizziness or vertigo, loss of balance and should be used with caution in activities requiring mental alertness (drivers, pilots, machinery operators, others).
- Sulfadoxine-pyrimethamine is not recommended due to worldwide resistance and it is not available in the U.S.
- All short-acting drugs – artemisinins, quinine – are not used for prophylaxis. Because of their short half-lives, they do not provide adequate protection against infection of the red blood cells by the parasites.

3. What antimalarial drugs would be appropriate for Andy?

- Atovaquone-proguanil or
- Doxycycline

Check your knowledge

4. JP is 38 years old and planning a 10-day trip to Asia where the predominant malarial parasite is *P. vivax*. JP is healthy with no medical conditions. NKDA. A prescription is provided for an antimalarial drug for prophylaxis of parasitic infection of the red cells. What is the drug? What are the directions for taking it? Why?

- Chloroquine is recommended for prophylaxis and treatment of non-falciparum malaria.
- Chloroquine is taken 1 to 2 weeks before traveling, then weekly during the duration of the stay, then for 4 weeks after returning.
- Chloroquine has an initial half-life of 3-5 days and a terminal half-life of 1-2 months. Chloroquine is sequestered in tissues: liver, spleen, kidney, lung, brain, spinal cord, melanocytes.

5. When JP returns to the U.S., he is provided a prescription for a long-acting drug taken as a single dose and directed to continue the antimalarial drug he was taking during his stay once weekly for an additional four weeks. What is the purpose of the additional drug? What are two contraindications to its use?

- Tafenoquine is prescribed to eradicate the *P. vivax* hypnozoites and prevent relapse.
- Tafenoquine is contraindicated in people with glucose-6-phosphate dehydrogenase (G6P deficiency (<70% or normal activity).
- G6PD screening must be performed before prescribing tafenoquine or primaquine.
- Tafenoquine is also not approved for in pediatric patients (<18yo, prophylaxis; <16yo treatment.)

Check your knowledge

6. AK is a 25-year-old woman invited to a family wedding in West Bengal in one month. She was born in the U.S. and is pregnant at 23 weeks gestation (G1,P0). She states that she is excited to see her family, whom she has not visited in ten years. For malaria prophylaxis, which of the following is the best option for her?

   A. Atovaquone-proguanil
   B. Chloroquine
   C. Mefloquine
   D. Doxycycline
   E. Do not travel to South Asia

- Option E: Don’t go
- Malaria infection in pregnancy is a major cause of maternal death, maternal anemia, miscarriage, preterm birth, fetal growth restriction, low birth weight, stillbirth, congenital infection, neonatal mortality.
- Pregnancy increases the chances of developing severe malaria and severe disease.
- *P. falciparum* infected red cells can sequester in the in the placenta and cause adverse pregnancy outcome.

Check your knowledge

7. Angelica is a previously healthy 4-year-old female presents to the ED with fever and chills, rapid heart rate, rapid breathing rate, headache, fatigue, nausea, diarrhea, and joint and muscle pains for 24 hours. Signs and symptoms of severe disease are absent. History is significant for recent travel to Haiti. Physical exam and laboratory findings are consistent with *P. falciparum* infection. The patient is able to take oral medication. Which of the following is the first-line therapy for this patient?

A. Artemisinin-lumefantrine
B. Artesunate
C. Chloroquine
D. Quinine-doxycycline
E. Primaquine

- Artemisinin-lumefantrine is the preferred drug. The tablets can be crushed and mixed in water if the patient cannot swallow pills.
- Artesunate is given IV or IM for treatment of severe malaria until the patient can take oral meds.
- Chloroquine is an alternative for sensitive *Plasmodium* infection. There are reports of chloroquine resistance in Haiti so ACT is a better option to be sure of effectiveness in curing malaria.
- Quinine would be effective but is not preferred because it can cause hypoglycemia and blackwater fever (intravascular hemolysis hemoglobinuria).
- Doxycycline is not given to patients <8 years old because it can cause discoloration of tooth enamel. Clindamycin would be given instead.
- Primaquine is for eradication of *P. vivax* and *P. ovale* hypnozoites for “radical cure”.

References

- Access Medicine Goodman & Gilman The Pharmacological Basis of Therapeutics 14e, 2023; Chapter 66: Chemotherapy of Malaria
- Access Medicine Harrison's Principles of Internal Medicine, 20e, 2018: Chapter 219: Malaria
- Access Medicine Jawetz, Melnick, and Adelberg's Medical Microbiology 27e, 2016; Chapter 46: Medical Parasitology
- Access Medicine Katzung's Basic & Clinical Pharmacology 16e, 2024; Chapter 52: Antiprotozoal Drugs
- Access Medicine Sherris Medical Microbiology, 7e, 2018; Chapter 51: Apicomplexa and Microsporidia
- Centers for Disease Control and Prevention > Parasites > Malaria https://cdc.gov/parasites/malaria/index.html
- Lexicomp monographs
- UpToDate various topics, literature reviews are current
- WHO Guidelines for the Treatment of Malaria, 30 November 2024

https://www.who.int/teams/global-malaria-programme/guidelines-for-malaria

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